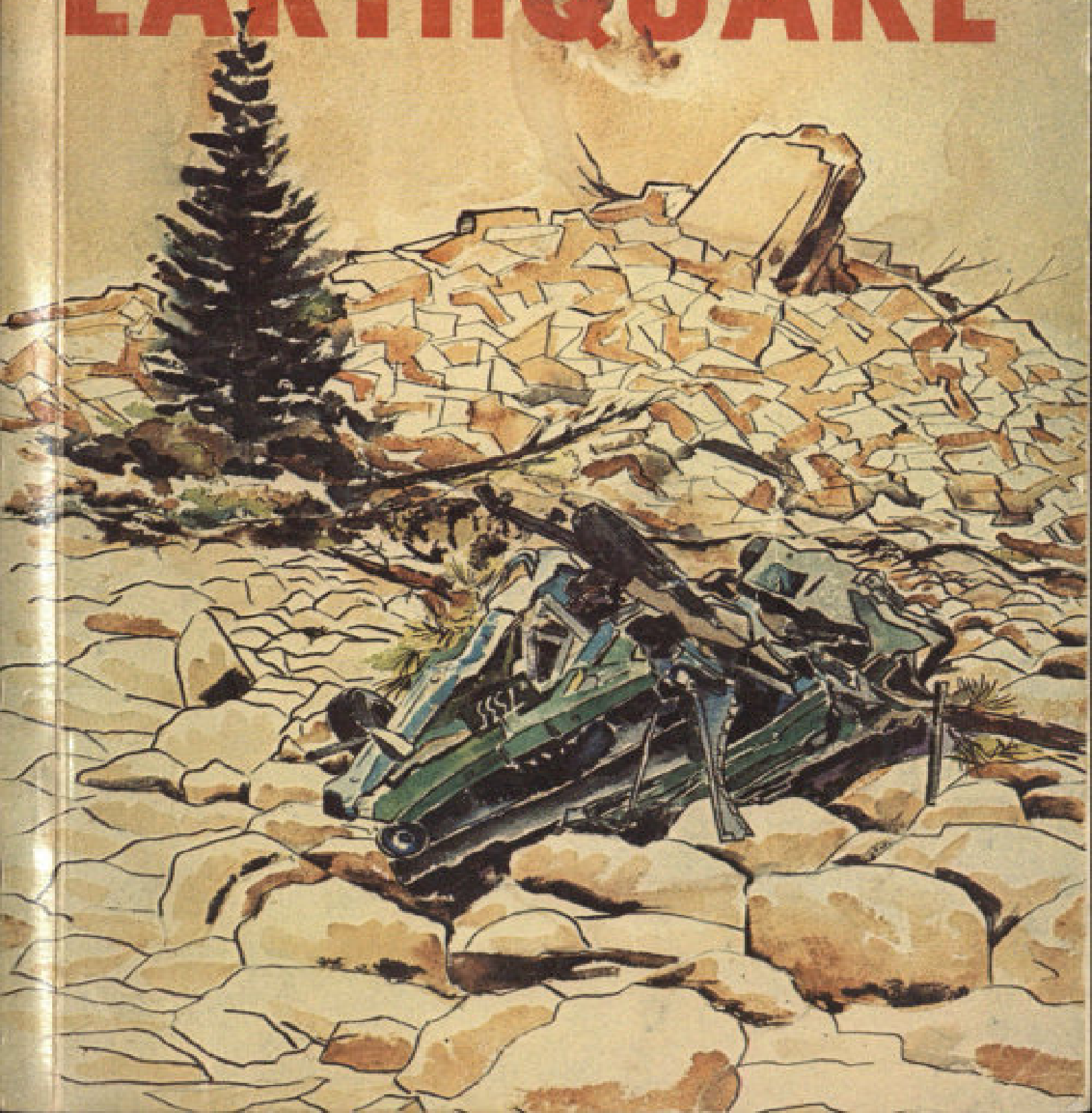


The Night The Mountain Fell

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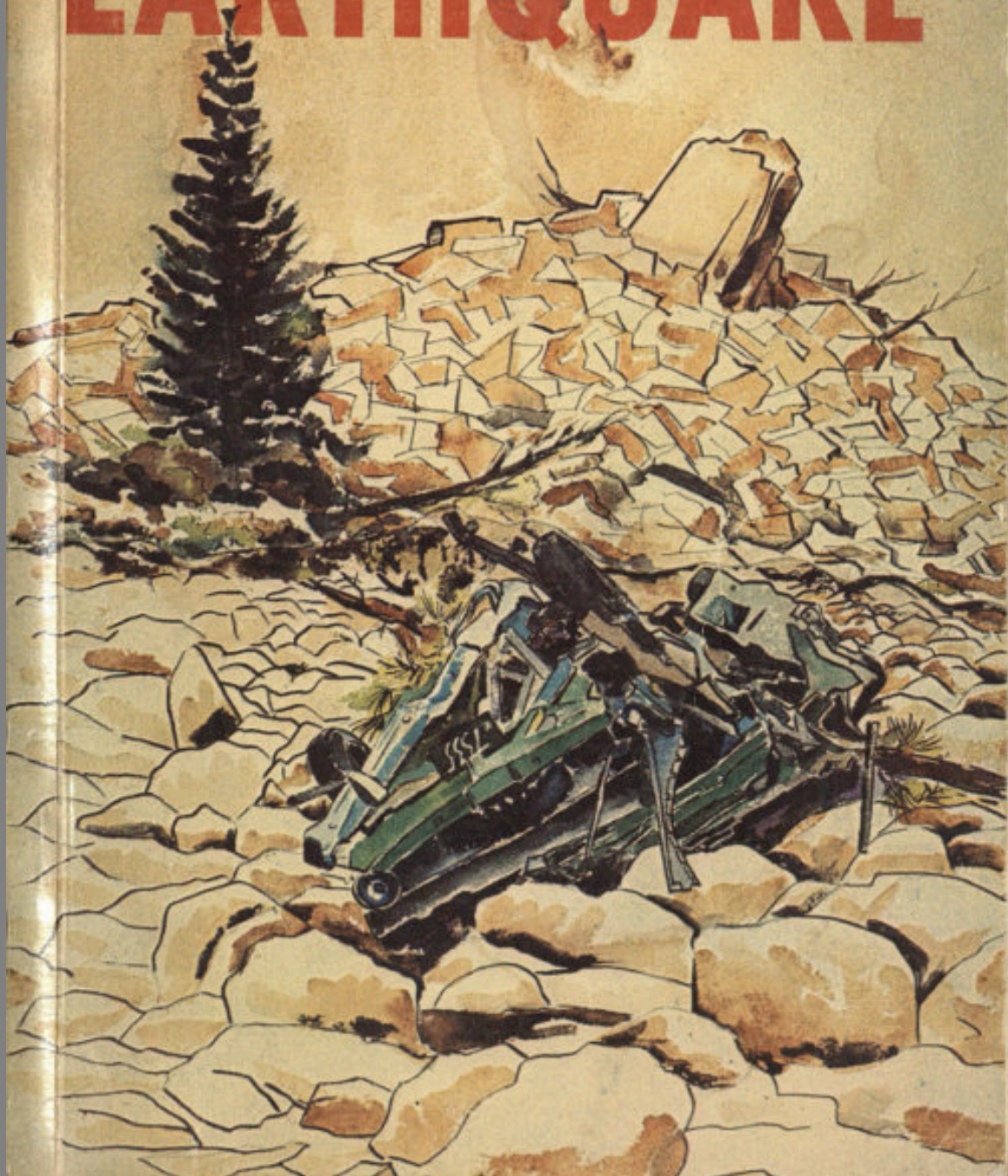
MONTANA-YELLOWSTONE EARTHQUAKE



The Night The Mountain Fell

THE STORY OF THE

MONTANA-YELLOWSTONE EARTHQUAKE



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EARTHQUAKE ***

THE NIGHT THE MOUNTAIN FELL

THE STORY OF THE

MONTANA-YELLOWSTONE EARTHQUAKE

—by *Edmund Christopherson*



Cover by Elwood Averill

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CONTENTS

<u>Foreword and Dedication</u>	3
<u>Real Shook</u>	5
<u>Trapped</u>	16
<u>Outside World</u>	26
<u>First Patrol</u>	32
<u>CD's Puzzle</u>	37
<u>Rescue—first MD</u>	39
<u>CD Wrapup</u>	48
<u>Untrapped</u>	54
<u>Mystery—Who Got It?</u>	60
<u>CD and Natural Disasters</u>	67
<u>Aftermath</u>	72
<u>Living Geology</u>	75
<u>Now You Can See</u>	84

FOREWORD AND DEDICATION

*To those who experienced, suffered
the helping and the helped, the surviving and the lost
all members of the involuntary fellowship
of the Montana-Yellowstone Earthquake,
to those who come to see and wonder,
and especially to those who assisted in this book's realization,
"The Night the Mountain Fell" is cordially dedicated.*





80 million tons of rock crashed off the right wall, blocking the mouth of the Madison Canyon.
(Christopherson)



Where the mountain fell. The tremendous slide tumbled off snow-chuted sections of mountain in center.
(Montana Power Company)

REAL SHOOK



August is a busy month in the exciting mountain vacation area that centers in West Yellowstone, Montana, and includes Yellowstone National Park, the restored ghost town of Virginia City, the nationally famous trout fishing reach of Madison Canyon that runs through the Gallatin National Forest, plus dude ranches and lakes in the parts of Montana, Wyoming and Idaho where the three states come together.

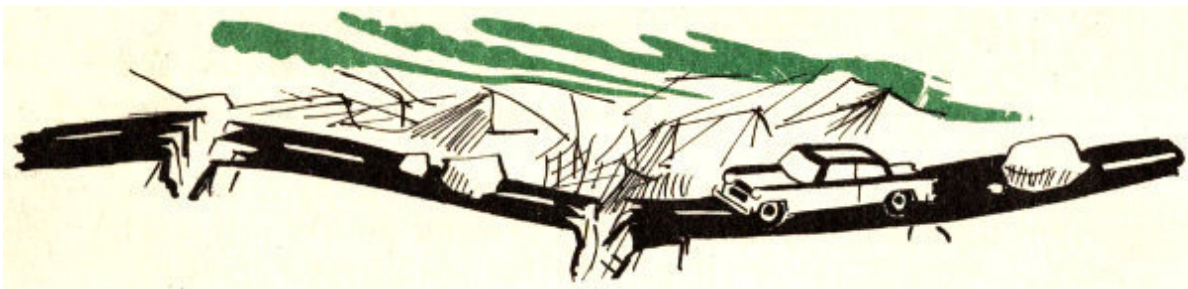
Geologically, it's a new area, where enormous forces are still thrusting up mountains, where volcanic craters still exist, and where the heat of the earth still spouts its imprisoned fury through the geysers that have made Yellowstone Park's Firehole Basin famous.

At 11:37 P. M. on Monday, August 17, 1959, one of the severest earthquakes recorded on the North American continent shook this area. It sent gigantic tidal waves surging down the 7-mile length of Hebgen Lake, throwing an

enormous quantity of water over the top of Hebgen Dam, the way you can slosh water out of a dishpan, still keeping it upright. This water—described as a wall 20 ft. high—swept down the narrow Madison Canyon, full of campers and vacationers who were staying in dude ranches and at three Forest Service campgrounds along the seven-mile stretch from the dam to the point where the canyon opened up into rolling wheat and grazing land. Just about the time this surge of water reached the mouth of the canyon, half of a 7,600-ft.-high mountain came crashing down into the valley and cascaded, like water, up the opposite canyon wall, hurtling house-size quartzite and dolomite boulders onto the lower portion of Rock Creek Campground.

This slide dammed the river and forced the surging water—carrying 6 trees, mud, and debris, back into the campground. The campers who'd escaped being crushed under part of the 44 million cubic yards (80 million tons) of rock found themselves picked up and thrown against trees, cars, trailers, the side of the canyon, etc. Heavy, 4,000 pound cars were tossed 40 ft. and smashed against trees by the force of the ricocheting water and the near-hurricane velocity wind created by the mountainfall. Other cars were scrunched to suitcase thickness and thrown out from under the slide.

And the water stayed—held by the earthquake-caused natural dam. It began to flood the lower end of the canyon. At the upper end, big sections of the road that would take the 300 people trapped in the canyon to safety crumpled and fell into Hebgen Lake, cutting them off from the world outside.



When the quake hit, summer Alternate Rangers Fred Tim and Lamont Herbold were on duty at the West Yellowstone entrance of Yellowstone

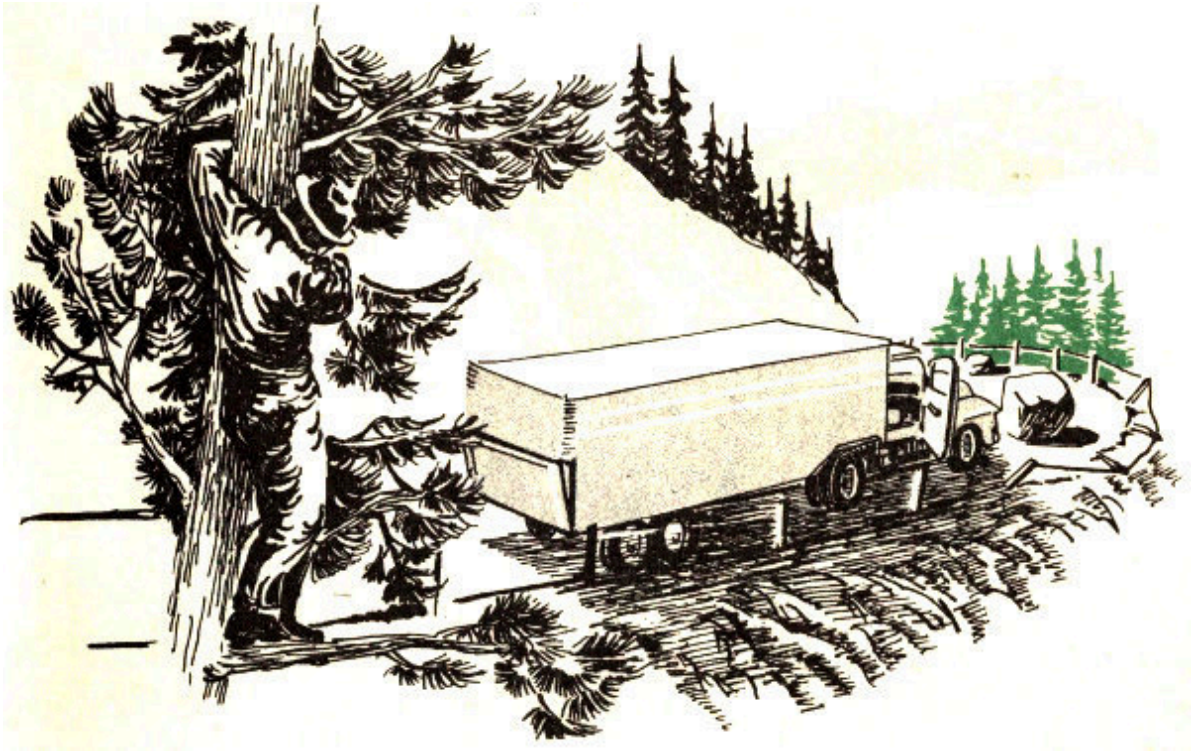
National Park. They had just cleared a semi-load of Pres-to-Logs. As the truck pulled on through the gate, the plywood gatehouse shook so violently, with the lights flashing off and on, that Herbold shouted,

“Stop the truck, you _____, you’ve hooked the shack!”

Truck drivers Jack and Lyle Tuttle thought the frantic way their truck was flopping around meant the motor had broken loose from the mounts. Driving into the Park, they were halted by huge rocks blocking the road. Renewed shaking, with tons more rocks rolling down the mountainside sent them scurrying for cover behind trees. Lyle took refuge in a tree, where, he later said, the shaking seemed twice as rough.

When the quaking stopped briefly, they turned the truck around and were happy to get out before more boulders blocked their exit.

In the confusion that followed when the first shock hit, Jerry Yetter, who operates the Duck Creek Cabins near West Yellowstone, jumped out of bed and knocked on all the cabin doors to warn the occupants of the quake. Only after he’d finished the job did he realize that he was wearing no clothes at all.



His wife, Iris, ran onto the front porch. The porch dropped into the basement. She climbed out, got into the car, and didn't stop until she reached Bozeman, 90 miles to the north.



Just west of the Duck Creek Junction of highways 1 and 191, the first shocks wakened Rolland Whitman as it sent dishes and furniture crashing to the floor. When he couldn't reach his wife's folks in West Yellowstone, 10 miles south, by phone, he rushed his wife, Margaret, and their six children into the car, started out, and immediately crashed over a 13-foot drop-off scarp that the quake had juttied up between his home and the highway.

On the night of the quake Mrs. Grace Miller, a widow who, in her 8 seventies, is still sprightly enough to run, single-handed, the Hillgard Fishing Lodge cabin and boat rentals on the north shore of Hebgen Lake, found herself suddenly wakened about midnight. She didn't know what was happening, but she felt she had to get out of the house. She threw a blanket around herself. The door was jammed, and she had to kick to get it open.



Outside the door she saw a big, 5-foot crevice. As she leaped across it, the house dropped from under her into the lake. More crevices kept opening in the moonlit ground as she walked away from the lake. “Rabbits were skedaddling in every which direction,” she said, but her Malamute dog, Sandy, was so frightened he wouldn’t even notice them.

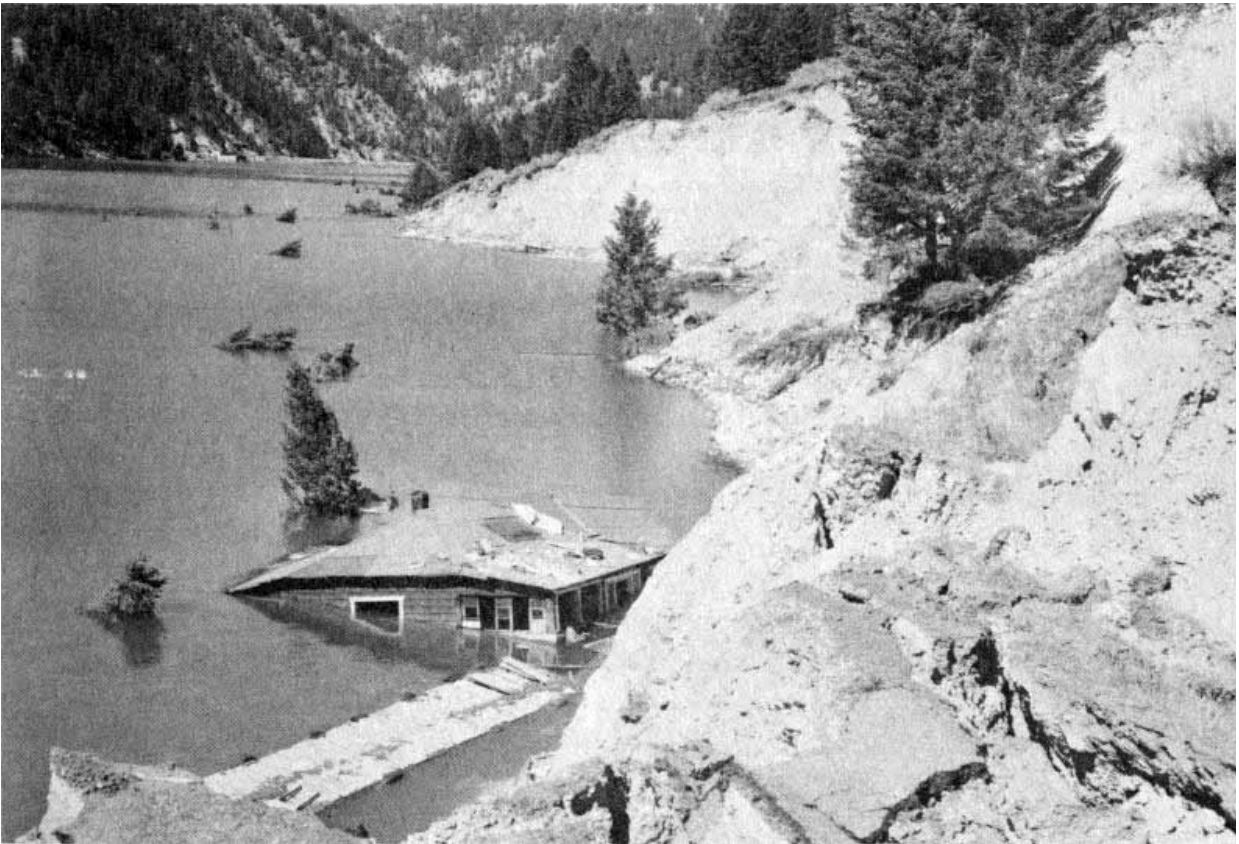
After quite a spell of hiking in the nightmare-like night, she found refuge along with about forty other people at Kirkwood Ranch, which itself was considerably damaged, but a safe distance from the lake. She was safe

there, while next day skin-divers, alerted by worried friends, searched her floating house for her body.

Later next day she boated past her 9-room home—which contained everything she owned, floating on the lake.

“I hope it stays upright,” she said. “My teeth are still on the kitchen counter, right next to the sink.”

When she arrived at the dam, she greeted an acquaintance with, “I’ve been a pretty tough old bird, but I wouldn’t want to go through that again!”



Mrs. Grace Miller’s house, which dropped into Hebgen Lake, floating along what used to be Route 287.
(U. S. Forest Service)

In a forest fire lookout on top of 10,300-ft.-high Mt. Holmes in Yellowstone Park, the first shock threw Penn State College student David Bittner out of his bunk.

“By golly, they’ll believe me this time,” he said with satisfaction as he picked himself up off the floor.

Several days earlier he’d phoned a report of substantial tremors, but no one would take his report seriously.

Charles Godkin, chef at the Frontier, and his wife, Ruth, a waitress, 10 were driving home at 11:37.

“We must have a flat,” she said as the car thumped and shook along the road.

When Godkin got out to look, the ground was bucking so strenuously that he could hardly stand up. Back at the Frontier, he found steak plates all over the floor. In the establishment’s walk-in freezer he found the floor covered with mayonnaise—a foot deep!



At the Emmett J. Culligan place, dubbed the “Blarneystone Ranch,” the Santa Barbara water softener tycoon spent hundreds of thousands of dollars building a refuge from the possibility of atomic attack.

Ironically, the main fault of the earthquake rammed through one end of his building’s cement block foundation, raising the ground 15 ft., twisting and cracking the whole 150-ft. length of the building.

Ironically, too, Culligan’s spread was perhaps the only one reputed to be covered by earthquake insurance.

His caretaking family, John and Doris Russell, were trapped in their cottage and had to crawl out and pass their children through a chin-high 15-inch square window.



At the proud dude ranch, Parade Rest, where Bud and Lu Morris capitalize on the area’s superb fishing, the shock toppled chimneys atop the massive log buildings and sent the guests scurrying outdoors.

Huddled around a huge campfire in the courtyard, where it seemed safer, they felt bewildered and helpless as the ground continued to heave and writhe throughout the night. For hours, the shocks continued at the rate of one every minute.

By morning the kitchen was a shambles—“like a cabin a grizzly bear had worked over. Dishes, flour—everything crashed to the floor. The only thing to do was to clean it up with a broom and shovel,” Lu Morris said.

11



The fault scarp running through the Culligan place.

(U. S. Geological Survey)

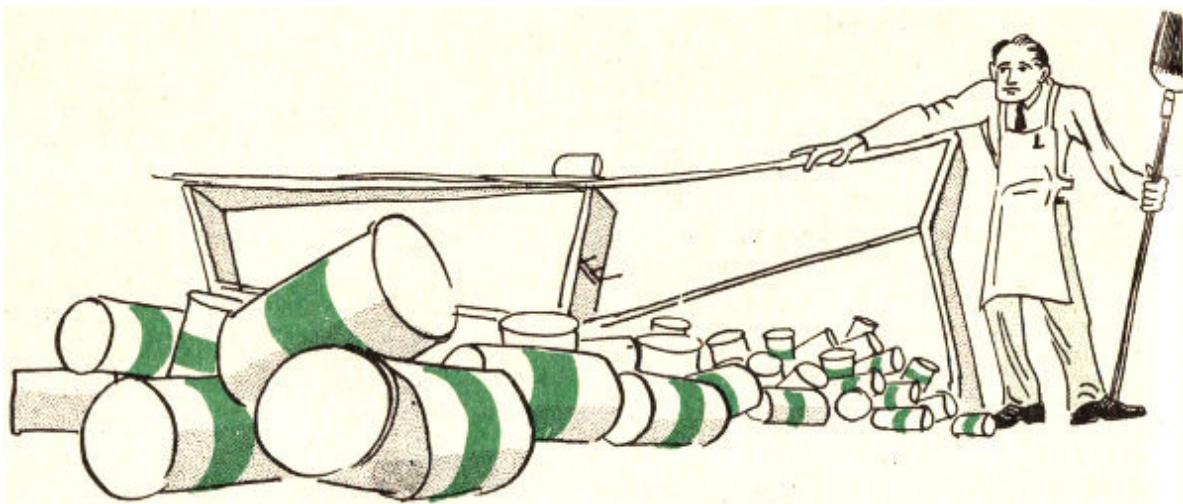


This ground was level before the quake.

(Christopherson)

Elsewhere throughout the earthquake area, crockery and goods in glass containers were at a premium; drug stores, bars, groceries were shard-piled shambles.

12



After the quake, the proprietor of the antique shop next to the West Yellowstone Post Office took one look at the disheartening spectacle of his shop and took off. The shop floor was strewn with a fortune in broken antique glass and dishware.

“The ground just got up and bucked like a horse,” one West Yellowstone citizen put it.

The only man who was enthusiastic about the earthquake from the start was geologist Irving J. Witkind of the U. S. Geological Survey, who was living in a trailer on a rise to the north of Hebgen Lake, above the Culligans and Parade Rest, while he surveyed and mapped the area.

When the first shock hit, he figured his trailer had somehow broken loose and was rolling down the hill. He charged out, intent on stopping it. From the way the trees were swaying in the absence of any wind, he knew it was a genuine earthquake. He hopped in his jeep and headed down toward the lake. He saw the scarp that the Whitmans soared off just in time to stop.

“It’s mine! It’s mine!” he shouted as he got out of the jeep and realized the full measure of his fortune. His words will echo wherever geologists gather in years to come. Professionally, his once-in-a-thousand-lifetimes fortune in being on the scene of a major quake meant as much as discovering an unfound Pharaoh’s tomb would to an Egyptologist.

At Mammoth, the old army post which is still headquarters for 13 Yellowstone Park, Superintendent Lon Garrison was sitting up in bed reading when the quake hit. His wife and daughter were watching TV when the big chimneys and rocks from the massive old 1909-built masonry buildings began crashing through the porches and roofs.

“We got out and fast. We prided ourselves on being cool. It wasn’t for an hour or so that I remembered that I was still wearing my Park Service uniform coat over pajama pants.”

Every time there was a new tremor, the coyotes, abundant thereabouts, would let out a fresh howl. The phone lines to Old Faithful and West Yellowstone weren’t working. The quake had taken them out. The 18,000

people who were overnighting in the Park when the quakes began were on the edge of panic.

“What can we do?”

“How can we get word out?”

“Can we get out?”



At Golden Gate, south of Mammoth, in Yellowstone Park, the road was blocked by this deluge of rock loosened and shaken down by the quake. (National Park Service)

Everyone wanted answers to these questions at once.

14

At Old Faithful, 800 people were in the recreation hall enjoying a college talent program. In the best entertainment tradition, the MC played it cool, continuing his patter while the Park Rangers opened the doors. Everyone exited in good order.

But there was to be little comfort that night. Everyone who'd made it to bed got up after the first shock.

At the massive, log-built Old Faithful Inn, the timbers gave out loud creaking and popping noises as the structural torment continued.

"We had to evacuate the building," Superintendent Garrison said. "Hot water from a broken pipe in the attic was running down the floor of the east wing. Half an hour later the fireplace and chimney crashed through the dining room floor, activating the sprinkler system. The water damage was horrible.

"A few hours earlier, with the dining room full, the casualty list would have been gruesome.

"As it was, our only casualty was a woman who sprained her ankle leaping out of bed after the first tremor.

"Later in the week a ranger, exhausted from quake duty, skidded on a rain-slick pavement and went off the road.

"We feel that God had his arm around us all the way."

The quakes continued with especial violence at Old Faithful. Evacuees from the Inn sat out the night, wrapped in hotel blankets, in their cars and in the big, distinctive Yellowstone Park Co. busses, trembling with fear at each new quake.

At the new Canyon Village, guests were reassured by the big-voiced man who, in the midst of the turmoil, marched up to the reservations desk and demanded accommodations for an additional two nights.

Canyon, too, was the place where, they say, another guest left a note on his pillow for the chambermaid, saying,

"An awfully rough bear stayed under my cabin last night. Had an awfully hard time sleeping. Better tell the night man to do something about it."



The force of the quake shook this retaining wall loose—threatening the Gibbon Falls Road in Yellowstone Park.
(National Park Service)

As the shocks continued, the summons to exodus was clear. Quake-broken roads blocked all the exits from West Yellowstone except the route, 191, through Idaho south to Pocatello. For the rest of the night it was bright with the lights of cars streaming away from the earthquake country to the solid security and comfort of the outside world.

TRAPPED

For trailer and tent campers, attractive Rock Creek Campground, less than a mile from the mouth of Madison Canyon, was a favored site. So much so that it was full most of the summer season. Campers who pulled into the canyon too late to find campsites in the formal, or improved, area just pulled off the road and overnighted on any level spot they could find along the road.

Two vacationing families, the Osts and the Fredericks, felt lucky when they found adjacent campsites at Rock Creek on Monday, August 17. Rev. Elmer Ost, who teaches psychology at Biblical Seminary in New York City and doubles as pastor at Bethany Congregational Church in Corona, Queens, his wife, Ruth, youngsters, Larry, 14, Geraldine, 13, Joan, 11, and Shirley, 6 had been enjoying a leisurely camping vacation in the Northwest.

The Melvin Fredericks family (he's a biscuit salesman for B & B from Elyria, Ohio) included Mrs. Laura, Melva, 16, Paul, 15, and George Whitmore, 15, who lives with the Fredericks in Elyria, Ohio, while his folks are missionarying in Brazil.

The two families met in Columbia Falls, Montana, at the home of Rev. Ralph Werner, who was a relative of the Fredericks and a college chum of the Osts. They'd both toured Glacier National Park and were headed for Yellowstone Park and the Black Hills, and decided to camp together the next night. Ost told Fredericks that if you didn't make it to Yellowstone Park before noon, the campsites would all be filled. But, he knew of camping areas in the Madison Canyon, near West Yellowstone, and not too far outside the park where they'd be more likely to find room. The Osts got to Rock Creek Campground at 6 o'clock Monday, August 17, found a site, and stationed Larry by the road to stop the Fredericks.

The two families chose a small open area near the stream and pleasant evergreen trees at the east end of the camp near the entrance.



The Elmer Ost family were special guests at the Madison County Fair at Twin Bridges, five days after their earthquake ordeal. (Christopherson)

They set up camp, ate together, socialized, and made plans to get up at 6 o'clock next morning, breakfast on dry cereal, and get an early start for Yellowstone. They swapped stories about their vacations, joked about the bear that was supposed to be scavenging around the campground, and decided to walk down to the highway where they'd be away from the trees and able to see the moon, which lit the mountain behind them. But the mountain that looked high on the south side of the canyon kept them from seeing the moon directly.

They turned in early—at 9 o'clock. The two younger Ost girls, mildly concerned about the bear, decided to sleep in the car, a 1950 Buick. Larry and Geraldine and their parents all slept in their tent. Mrs. Fredericks and Melva slept in their station wagon, while the men stretched out in a tent.

Everyone was nicely settled by 11:37, when a thunder-like commotion outside awakened them. Ruth and Gerry Ost shouted something about bears

as they jumped out of the tent.

“It’s a cyclone!” Mrs. Ost was screaming in half-wakened terror as Rev. Ost emerged from the tent.

The sky was clear, the moon bright, as Rev. Ost looked up and to the west. There were no clouds or wind.

But terror ran through the whole party as they saw tents swaying, trees shaking as though torn in a violent wind. Then the Osts’ 1950 Buick began to rock from front to rear as if men were jumping energetically on the bumpers. The brake lights went on as one of the gals jumped on the brake. 18

Then came a tremendous roar, like several express trains passing right through the camp. The trees shut off their view of the huge, 7,600 ft. mountain falling, of the huge boulders, big as houses, hurtling down one side of the canyon and up the other side, a mile away, throwing sparks and dust as they fell. Rev. Ost sensed the rushing of the wind and water trapped by the avalanching mountain, and thrown back at the dazed campers—at tornado speeds—from under the slide. “Hang onto a tree!” he shouted. Mrs. Ost ran for the car as she saw the wave of water coming. Larry was caught in the tent when the wall of water, mud, and trees hit them with such violence that it crumpled trailers and hurled the Osts’ 4,000 pound car thirty feet and smashed it against a row of trees. Although Mrs. Ost was holding onto the steering wheel, the violence of the surging water threw her against the side of the car so violently that it made a pulp out of the right side of her face. In the midst of the mud, water, and floating and flying debris, Larry managed to tear his way out of the tent.

Dust from the slide obscured the moon and heightened the sense of tragedy and terror.

The tent was gone. The deluge of water had jammed cars and trailers together. The rocks had covered the site of a trailer where a family had been playing earlier in the day.



The night rang out with the bewildered crying out for lost relatives. Stunned like the others, Rev. Ost shouted for “Fran.” After hitting the brake pedal she’d jumped out of the car and scampered, like a deer, to higher ground.

He found Gerry, unharmed, except for being wet and an injury to her hand. Sloshing through water to his knees, he found Ruth still in the car. After several minutes hunting and shouting “Larry” he found his son, soaked, clad only in shorts. 19



Mel Fredericks, lifting Rev. Elmer Ost, shows how he tried to free his trapped son, Paul.

(Montana Highway Commission)

The screams of the lost and losing continued. A woman handed a baby to Melva Fredericks, saying, “Comfort him.” One dazed man walked around crying out for his missing wife. From the wet and dark came the cry of another woman calling out, “It’s safe here!”—hoping to attract someone to help or keep her company.

The Ost women and Mrs. Fredericks struggled to higher, safer ground. When the Fredericks men didn’t show up, Ost left the women praying while he went to look for them. He heard Mr. Fredericks call for help for his son, Paul. With a flashlight he borrowed, he was able to see the difficulty. The surge of water and trees had caught 15-year-old Paul and pinioned him in a sitting position in the water, with one log across the small of his back and another across his lap. The ends of the log were jammed between a smashed trailer and the Ost car, so solidly they wouldn’t budge.

Paul cried out with pain as the two men tried to pull him loose. The water kept rising as the men tried to pry the logs apart with sticks. A 2 × 12 plank, ten feet long, even though full of spikes, seemed a promising tool to pry with, but with it they were only able to gain an inch or so further separation of the logs that held Paul prisoner. 20

The men felt Paul's and their own helpless panic as the water swelled up to his chest, his neck, his chin. Raised in a soundly religious family, Paul bravely faced the realization that he was gasps from death. In desperation, Ost called on Mel Fredericks to pull as hard as he could, not to care if Paul cried, or if they pulled his arms or legs out of joint. In this last, desperate straining try they found that miraculously they could raise him six inches. The rising water had buoyed the trailer. In their next few feverish tries they were able to pull him loose and helped him to walk to high, dry ground.

One stranded group, calling for help, included a wheelchair case, and, mucking shoeless through the water, they portaged him out, chair and all. George Whitmore had a badly injured eye—from running into a rope, and it looked like he might lose its sight.

They all moved to the highway, which was still dry. A motley crew they were, in pajamas, or almost unclothed, some shoeless. By this time the water had covered their cars. Some of the wounded were taken by car toward Hebgen Dam—away from the slide.

Marooned, without their cars, in a strange, shaking canyon—prisoners of a night in which everything seemed mad, somehow word reached them that their ordeal might not be over. There was possibility that a dam several miles upstream, which they'd never seen, was likely to give way any minute. So they scrambled up the sagebrush hillside and built a fire on a level and fairly open site.

Others joined the two families. One group, whose car hadn't been flooded so suddenly, managed to save groceries, a camp stove, sleeping bags, pans and a 9 × 12 ft. plastic tarp.

Without worrying about modesty, they dried themselves around the fire. There were 17 in the party. It cleared, and then clouds obscured the moon.

The ground kept shaking. With almost every new tremor came sparks and puffs of dust, and the terrifying, crashing echoes of another avalanche across the valley, and the realization that the valley side above them might go any time. The air was full of dust and the sickening smell of mud and torn fir trees. All through the night they heard the haunting cries of “Help, help—we’re freezing,” from the Grover Maults, who’d been marooned on top of their trailer and by this time were hanging onto a tree. 21

They worried about forest fires, and sang hymns to keep up their courage.

At 3:00 A. M. there was a thunderstorm and a light, continuing rain. They huddled under the plastic tarp—all 21 of them—and wondered what would happen next.



An elderly couple, the Grover C. Maults, a 72-year-old retired decorator and his wife, Lillian, 68, of Temple City Calif., had parked their trailer at the scenic Rock Creek Campground for a week before the especially beautiful, bright moonlight night of August 17th. There were lots of bears in the area, and like many other campers, when the first jolt hit them, they figured that the bears were trying to get into their trailer.

“No, it must be an earthquake,” Mrs. Mault said. 22

Looking out through the trailer window, the moon made it seem like daylight. Everything was going upside down. An instant later their trailer

was tossed end over end, landing, miraculously, on its wheels. Then it seemed as though something picked the trailer up and hurled it into the water.

Mault got his “nightie-clad missus” out of the trailer and lifted her on top, and went back into the trailer to get sweaters or something. It suddenly turned dark. The moon disappeared in dust. The water had risen to Mault’s chin by the time he got out of the trailer.

By the time he’d crawled on the trailer roof, put on trousers, a shirt and sweater, and wrapped clothing around his wife’s legs, the water was beginning to cover the trailer roof and rising fast. They prayed that the trailer would drift toward a nearby tree. It did. The first branch broke as Mault grabbed it. He barely had time to get one arm around the tree and hold onto his wife with the other when the trailer was swept out from under them.

“It was horrible,” he says. “As I tried to pull the missus up the limbs kept breaking off. I tried to grab higher limbs and cling to the missus with my legs. The limbs still kept breaking off. Finally we found a limb that would hold.

“We were surrounded by deep water. Through the night we hollered and hollered for help. People tried to get to us with ropes, couldn’t reach us, and yelled that we should hang on, they were going for a boat.

“While we struggled to hold on, we could see the mountains sliding and falling every few minutes. There’d be a terrific roar, followed by more slides. I thought the world was coming to an end. It turned hazy, with thunder, lightning, then began to rain.

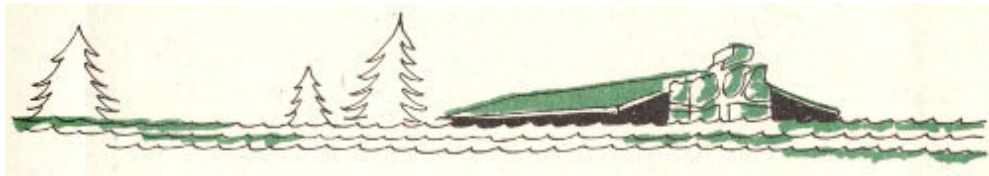
“As we clung to the tree, with water up to our necks, my wife slipped under three or four times. The last time she was gasping for breath, I managed to pull her out.

“‘Let me go and save yourself,’ she begged. ‘If you go, I’ll go, too,’ I told her.

“About 8:00 or 8:30 in the morning they came for us in a boat. It was just in time. We couldn’t have held out for another ten minutes. The water was

rising so fast that the rescuers had to move their truck three times before they could unload the boat. 23

“At first, when rescued, we could see lights, then everything went black. We couldn’t hear anything over the roar of the tumbling mountains. We were froze stiff from hanging on so long. We couldn’t move our legs. The men had to help us into the boat.”



In contrast to those who stood around and wondered was L. D. Smith, of Greeley, Colorado, who with his family was camped in a trailer at the Beaver Creek Campground a couple of miles downstream from Hebgen Dam. The loud noise and rumbling woke him. Outside the trailer, he found the water rising. The ground was shaking violently. He didn’t know what was causing it, but his first thought was that the dam had broken. The steep-walled canyon didn’t seem like a safe place for his family. As soon as the shaking subsided temporarily, he loaded his wife and two youngsters into the car and drove away from the dam, the collapse of which he instinctively felt was the greatest danger, as fast as he could.

A mile or two before he reached the slide, he ran into heavy dust. Still fearful of what the dam’s collapse would mean for those trapped in the deep canyon, when the slide blocked his path he turned off the road and drove up the north side of the canyon wall to a point where he couldn’t get any more traction. He then got his family out of his car and moved them to still higher ground.

Later in the evening his family joined the Osts and Fredericks around the fire on the hillside.

At about 8:00 o’clock on the same gorgeous moonlight night of August 17, the Purley Bennett family of Coeur d’Alene, Idaho, pulled their trailer off alongside the road on the flat at the mouth of the Madison Canyon 24

Gorge. They didn't plan to set up camp—just to rest for a few hours before continuing to the park. They talked a bit with others camped in the same informal area, then turned in. Purley, a 43-year-old truck driver, and his wife, Irene, slept in the trailer. The youngsters—Carole, 17; Phillip, 15; Tom, 10, and Susan, 5, stretched out in bedrolls outside on the ground.



“A car blew past, rolling over and over.”

(Christopherson)

They were awakened a little before midnight by a loud rumbling noise. They wondered what it was, but weren't concerned enough to get up or move their equipment. Some time later, in response to a much louder noise, Bennett left the trailer to see about the children. Mrs. Bennett was right behind him. As she stepped out of the trailer, she felt a strong wind coming up. There was a great rumbling, whooshing sound, and as the wind reached hurricane velocity she saw her husband grab a small tree for support. The wind swept him off his feet—he hung on like a flag tied to a mast. After a little bit he let go and was blown away. She never saw him again. She couldn't see 25 her children, except one flying through the air. A car was blown by, rolling over and over, and she found herself swept along with the trees, the rocks, and water.



The Bennett car—thrown out from under the huge slide onto the dried-up stream bed below.

(U. S. Forest Service)

“When I came to,” she said, “I was jammed against a tree with a log on my back. I don’t know how I got out. I thought I was the only one of my family still alive.”

Then, over the awful moaning of the boulders grinding and crashing and the sound of the tree trunks howling through the air, she heard the voice of her son, Phillip, calling.

Slowly, painfully, in spite of crippling injuries, they dragged, an inch at a time, toward one another over the rocky, oozy bed of the river which the huge slide had instantaneously stopped. Highway Patrolman Stevens, who found them several hours later, noticed how torn their hands were from this agonizing crawl.

That morning, in the hospital at Ennis, Mrs. Bennett told reporters, “They say my husband and my boys are dead, but I have faith. I know they will be found.”

They already had been—dead.

OUTSIDE WORLD

The Montana-Idaho, Wyoming area where the quake hit is a big, sprawled-out area where it's easy to get the feeling of isolation when everything's normal—the roads open, the phone lines and lights working.

In one shattering blow, the earthquake cut most of this area's access and communication. Big sections of the Yellowstone Park roads were blocked by slides and boulders. The road north of West Yellowstone was impassable. Big hunks of the road between the Duck Creek Y and Hebgen Dam had crumbled and slipped into the lake causing four major breaks and several minor ones. The big slide formed an 80 million ton block at the west exit of the canyon, and at Wade Lake, road breakup had immobilized another group of terrified campers.

For the first few hours after the quake, one of the biggest problems for the trapped was to get word out, and for those outside to get some idea of just what had happened.

At the instant of the quake, the Berkeley seismograph showed shock in the West Yellowstone area.

The first man to get word out was amateur radio operator Warren Russell, who operates station K7ICM from his trailer house in West Yellowstone, who began broadcasting news of the quake at 11:43 P. M.

At 11:50, another ham, Fr. Francis A. Peterson of St. Anthony, Idaho, contacted Idaho State Police, who relayed the word to HQ in Boise, and thence to the National Warning System at Battle Creek, Michigan.

At 12:25 A. M. on the first detail report from the Western Section of the Alert System it was reported that Hebgen Dam was demolished and that there were 6 feet of mud and water at the town of Ennis.



This upside-down Cadillac and the shook-up-road between West Yellowstone and the Duck Creek Y are typical of quake damage.
(Montana Highway Commission)





Several sections of road along the north shore of Hebgen Lake just slumped into the lake.
(U. S. Geological Survey)



(Montana Highway Commission)

When the quake hit at 11:37 P. M., it woke Austin Bailey, resident 29 maintenance man for the Montana Highway Department at Duck Creek Junction—where the road takes off along the north side of Hebgen Lake and through Madison Canyon. He noticed the light overhead jumping, furniture moved from the wall, the lights weren't working. Realizing that such a shaking would topple rocks onto the highway, he knew that he should get out and clear the roads before the heavy tourist traffic got underway next morning.

Outside everything seemed normal. He got into his own station wagon to make an initial check, started out, and 30 feet later drove over the 15 feet high scarp embankment—the main earthquake fault that had dropped off between the maintenance shed and the highway.

Shaken, but not hurt, he crawled out of the car, aware that something was seriously haywire, and that he had to call for outside help.

He went a mile to the nearest telephone. It was out. In the maintenance shed, where the heavy equipment and trucks are stored, he found the 16-ton rotary snowplow had been jolted eight feet out of its position the night before.

The radio transmitter in his pickup either wasn't working, or couldn't reach the area Highway Department HQ in Bozeman. The road, when he managed to reach it, was shredded by long cracks, running along the length of the road.

He loaded up his family and started north to get the word out that they needed help on the roads in the West Yellowstone area. At the Y he found an overturned Cadillac that had flipped coming over a continuation of the same scarp which ran through his own yard. Driving carefully—at times it was like straddling a grease rack—he finally found a phone that worked at Almart Lodge, 40 miles north of West Yellowstone. Highway District Engineer George Barrett logged Bailey's call at 1:50 A. M.

The quake caught Montana's Civil Defense Director Hugh K. Potter in bed. Potter, a grizzled former Montana Highway Patrol Captain and Helena Police Commissioner, had lived through Helena's 1935-6 earthquake. This earlier quake had logged some 3,000 recorded tremors, killed 4 people and destroyed several buildings, including Helena's City Hall. Potter wasn't greatly impressed by the somewhat diminished-by-distance initial shock, and went back to sleep.

At 1:30 A. M., the Helena Police roused out city fireman Ed Cottingham and reported that fragments of information about an earthquake which had caused severe damage in the state were coming in on police radio. At 1:32 Cottingham called Potter, and they went down and set up state Civil Defense HQ in Helena's City Hall, an Arabic-style former Shriners' building which also houses the capital city's fire and police department.



Standing on top the slide, with the fallen mountain as a backdrop, Civil Defense Director Hugh Potter and Madison County Sheriff Lloyd Brooks discuss problems created by the quake.

(Christopherson)

For the next two hours, their life was a turbulence seething with rumors. The steep walled canyons and high mountains which obstructed normal police short wave radio added to the problem of already disrupted communications in getting information out of the quake area.

Trying to piece together just what had happened, the damage, and what help was needed was like a horror movie about “THE THING,” with the exact nature of the horror emerging through the confusion and hysteria in small clues and fragments.

At CD HQ, Potter realized the possibility of Hebgen Dam’s collapse, bursting, shattering, breaking in the quake. It’s an un-reinforced

concrete-core earthfill dam 721 feet long, built in 1913 by the Montana Power Co. to regularize the flow of the Madison for downstream power generation. Its failure would threaten the tourists in the valley, and the sleepy 600 population town of Ennis, 65 miles below the canyon mouth.

Conflicting rumors filled the air—that the dam was destroyed. By 2:00 A. M., the police and Highway Department radio frequencies were zinging with these and other unconfirmed reports leaking in about the plight of the dam and the canyon. In Helena, Potter struggled with a vision of a smaller reenactment of the Johnstown Flood in Ennis if the dam did, or had, let go.



(Montana Highway Commission)

FIRST PATROL

Montana Highway Patrolman Glen Stevens made the first probe up the Madison Valley after the quake. In response to a request for help from Madison County Sheriff Lloyd Brooks in Virginia City, Stevens and Deputy Sheriff “Dutch” Buhl wheeled down to Ennis, arriving at about 2:30 o’clock. The telephone lines were out. It seemed important to warn people farther up the valley of the danger they were in. Some of the folks had already fled. One ranch family was still in bed. There were three groups of sleeping campers. They didn’t argue or waste time. When Stevens suggested they get out, they just left.

As Stevens and Buhl proceeded up the valley, they radioed in at frequent intervals that everything seemed normal. They reported rock on the road at various intervals from 26-mile hill on south to the place above Hutchins Bridge, where boulders tumbling from a rock cliff made the road impassable. Cabin camp operator Otto Kirby had got his people out, but there were two house trailers parked farther up, near the river. Stevens warned the occupants, and got them started out. They gassed up at Kirby’s ranch. It was cloudy and dark. At 3:15 o’clock Stevens radioed in that the water was muddy but otherwise seemed OK, and that he planned to cross the bridge and drive up along the river on an old road on the south side. Sheriff Brooks tried to discourage them, shouting via radio—“Don’t do it, you crazy bastards, the dam’s broke, and you’ll get killed too. Come back!”

It was this message which, picked up on other radios, and relayed to Helena, sparked CD director Hugh Potter to order the evacuation of Ennis.

With Sheriff Brooks’s warning fresh in mind, Stevens says, “Every turn we got off that bench, I thought we were going to meet swimming water.”



These huge boulders crashed down onto Cliff Lake Campground killing the Lloyd Strykers of San Mateo, California, without disturbing the food on their campsite table. (U. S. Forest Service)

As they moved up the valley, they got the message that a couple of people had been killed in the campground at Cliff Lake, to the south of the Madison, so when they hit the Raynolds Pass road, they headed that way.

They got there at daybreak, about 4:45 A. M. They found that a rock cliff had fallen across the road which ran along the lake, marooning the campers.

At the campground they found two campers dead, killed in a bizarre and gruesome accident. The E. H. Strykers of San Mateo, California, were camped on an improved campsite, with a fire site, a picnic table, and a place to park their car. Their three youngsters slept in a tent 100 feet away. The quake dislodged huge, eight-ton chunks of rock, and set them bounding downwards in a freakish, crescent shaped path, tearing the ground, and

toppling 60-year-old fir trees in their downward rush. Nimbly, two of these boulders bounded over the picnic table, stacked with food, and landed squarely on top of the sleeping bags in which the Stryker parents slept.

34



Cliff Lake tragedy—another view.

(U. S. Forest Service)

“It wasn’t pretty,” Stevens said. “But there wasn’t anything we could do. The rocks were too big to move, so we went down toward the Shaw Ranch and got Frank Shaw to take his 4×4 truck up and move the rocks off them.

“We drove back down to the highway to continue up to the canyon. In the freshening daylight on the way down from the high bench back into the valley we could see a couple of trailers down the highway near the mouth of the canyon. A Fish and Game Commission plane flew over, radioing something about an obstruction across the lower end of the canyon, and having two to three hours to get the people out. We had no idea what they meant. We got to the trailers. They wanted to know how to get out—the next section of the highway was blocked with rocks and boulders. We routed them across the river and out Raynolds Pass way into Idaho.

“One of the guys said he thought there were a couple of people still alive across the river. We got to the slide about 5:45 o’clock—the huge pile of millions of tons of rock where the highway used to be. You couldn’t believe what you were looking at. 35

“Somebody said something about a ‘little slide’.

“‘Little!’ I said to Dutch, ‘I’d hate like hell to see a big one!’

“The after-shocks that kept happening—with rocks crashing down and dirt and dust blowing up—didn’t contribute to our peace of mind, either. But we didn’t have much time to look.

“We struggled across the river—the slide had stopped the water, but left a muddy ooze and some water lying around in pools as much as three feet deep. We found Mrs. Irene Bennett lying in the rocky stream bed. She was cold and shivering. She didn’t have a stitch on. Neither did her son, Phillip, who was lying near her. Both of them were bruised and bashed. The Bennett boy had a broken right leg, shoulder, etc. We put Mrs. Bennett on an old wooden frame canvas cot and started across the slippery river bed with her. She must have weighed 180. As we struggled through the slippery muck she kept apologizing for causing us so much trouble, and told us about her husband and three other children, the other folks who were camped near her, and the tremendous spurt of wind and mud that threw them out from under the slide. She told how she’d come to—believing herself the only one of her family who’d survived. Despairing, she heard Phil calling from a spot 75 feet away where the water had thrown him. Their torn hands gave the story

of the agonizing effort these crippled survivors had made to drag themselves together over the rocky stream bed.

“By radio we asked the Fish and Game Commission plane flying overhead to go to Ennis and get Dr. Losee and fly him back, and land him on the highway nearby. We didn’t want to disturb Mrs. Bennett by moving her off the cot, so we put her into a station wagon. Morris Staggers, who lives nearby, showed up with an old iron bedstead, older than anyone there, and heavier, too. I’ll never forget the struggle we had carrying the Bennett boy across on it.



Rescuers prepare to move body of T. Mark Stowe of Sandy, Utah, who perished as he was thrown out from under the slide. Note tremendous water damage and dry river bed.

(United Press International)

“We took the Bennetts up to where the plane landed on the highway and turned them over to the doctor. Returning to the slide area, the increasing light made the slide seem even more formidable. That morning, working with Fish & Game Commission, etc., we found all of the people Mrs. Bennett had told us about except one. Like Mrs. Bennett and her son, all of these bodies had been stripped of their clothing and showed the effects of being beat to hell by wind and water. The coroner said all five of them died by drowning. We never did find Mrs. Marilyn Stowe, wife of Sandy, Utah

elementary school music teacher T. Mark Stowe, whose body we did find—she must still be under the slide.

“I just don’t care to go through any more mornings like that,” Stevens said.

CD'S PUZZLE

At 2:00 A. M. Potter called in Montana State Highway Engineer Fred Quinnell, Don Brown of the Montana Fish and Game Department, and Captain Alex Stephenson, Chief of the Montana Highway Patrol, to help sort out the rumors in the tense hours ahead. At 2:15 A. M. George Barrett, highway engineer in Bozeman, radioed to Helena Austin Bailey's report on road conditions in the West Yellowstone area.

Another report from a road maintenance man in the Ennis area brought some word of a rise in the Madison, way downstream from the Canyon. When asked by phone, Jack Corette, president of the Montana Power Co., said that he felt it unlikely that the dam had gone out. As a precautionary measure to protect communities farther downstream on the Madison, an immediate drawdown of Meadow Lake and Canyon Ferry reservoirs was begun.

At 2:53 Bozeman Sheriff Don Skeritt reported that through the ham radio net that was rapidly taking up the slack in communications, he'd messaged his deputy, Everett Biggs, in West Yellowstone. A short time later he'd received word that there was still much violent shaking in the area ... that the lake had gone down substantially ... and the dam was still holding.

Associated Press sent a man over to CD HQ in Helena at 2:55 A. M., to keep in touch with developments. At 3:15 o'clock, when out of the communications box came the cry, "It's gone! It's gone!" it was difficult to keep the press from rushing to the phones and announcing nationally that the dam had collapsed. There was no confirmation, but the impact of the moment impressed Potter to the extent that he immediately called the marshal, George Hibert at Ennis, the first settlement downstream from Hebgen Dam, and urged him to get the people out.

The sirens blasted 15 times. As one crusty old Ennis evacuee, Ray "Tuffy" Kohls, put it—"They wake you up in the middle of the g.d. night with the story that the dam's going to go,—still the people packed up

and got out in pretty good order. Of course there was some confusion. One guy grabbed a flashlight and a thermos of coffee. His wife got into the car wearing a coat over her nightgown, and carrying a girdle she'd been sewing on that evening."

Some of the evacuees drove over the hill to Madison County's exciting, historical county seat town of Virginia City to wait out the expected flood. But most Ennis folks spent the rest of the night perched in their parked cars on a hillside overlooking the town—like penitents waiting for Judgment Day.



Refugee family from Ennis stalk the streets of historically fabulous Virginia City the morning after the quake.
(Ken Lewis—Montana Standard)

RESCUE—FIRST MD

August 17 was the first time that Dr. Raymond G. Bayles, an active Bozeman MD, had got to bed at a decent hour in weeks. The tremors he felt in Bozeman were strong enough to damage buildings on Montana State College campus in Bozeman. Recently Bayles had bought the 50-room Stagecoach Inn at West Yellowstone, and he was concerned about the Inn and its employees. The phone service to “West” was out. As the night dragged on, the radio brought him the news that “West Yellowstone was close to the center of the quake, and that the road to ‘West’ was impassable.”



Road dropoff on Route 287 along north shore of Hebgen Lake between Hebgen Dam and the Duck Creek Y. (Wide World)

He chartered a plane at daybreak. On the way to “West,” he had the pilot fly down over the Madison Canyon. The dust had pretty much settled so they could see the massive slide in detail. Just above it a group of people (the Osts, Fredericks, Smiths, etc.) were waving for help. The lake was beginning to form in the canyon behind the slide. More people were standing near their cars and trailers halfway between the slide and the dam. Just below the dam was another caravan which included many station wagons. On the dam spillway someone had spelled OK-SOS with pancake flour in big white letters, and marked a big cross on the highway, in a spot suitable for a helicopter landing. Dr. Bayles realized that there were injured among those trapped in the canyon. 40

As they flew low over the lake, they saw where buildings and big sections of highway had dropped into the lake. On the usually clear surface of the lake, somehow as a result of the quake, thousands of logs appeared, probably submerged logs shaken off the bottom. There were virtually no boats visible.

At the Stagecoach Inn he found the staff huddled around a bonfire under the trees across the street, where they’d been since the heavy tremors began cracking plaster in the building. The exception was Jane Winton, a nurse, who managed the hotel, and had bravely stayed on duty at the desk.



Stranded docks raised by the quake action which raised the south shore of Hebgen Lake eight feet.
(U. S. Geological Survey)

They gathered splints, medicines—whatever emergency material they could find—and went to the airport. The pilot flew to a field big enough to land on at the Watkins Creek Ranch, on the south side of the lake about two miles from the dam.



In walking toward the dam, they found debris where the tidal wave had thrown it half a mile up from the shore. The entire south shore of the lake had risen about eight feet. After a mile's walking they came to a spot where people were dragging their boats higher out of the water. They believed the dam was going out, and at first didn't want to lend Dr. Bayles a boat, but he finally persuaded them. Dodging the many logs in the lake made the trip difficult. As they approached the dam, Jane Winton was frightened at the big crack in the dam's concrete core.

"Just you keep watching that crack," Dr. Bayles told her. "If it gets bigger, you'll know what's going to happen. If not, you'll be telling your grandchildren about all this."

"To reach the bank, we had to land through a lot of debris that had gathered at the dam. We went over the top of the concrete at 9:00," Dr. Bayles said.

“We were met by a girl who seemed to have more authority, Mildred (Mrs. Ramon) Greene of Billings, Montana, a former nurse who was one of the real heroes of the disaster. She told us that no one had been there, and that they’d had no word from outside since the quake, more than nine hours earlier.

“Mrs. Greene had the injured—there were sixteen serious cases—in the back of station wagons—two to a wagon, except for one elderly couple who were in their fishing trailer. Ray Painter, 46, a service station operator from Ogden, Utah, and his wife, Myrtle, 42, they were perhaps the most seriously injured. She had flesh torn off her arms, a crushed chest, a punctured lung, and hemorrhages from an arm artery. Her husband had deep lacerations over 90 per cent of his legs. Like the other injured, they were suffering terribly, yet not one of them was complaining.



Quake victim, Mrs. Margaret Holmes of Billings, Montana is given plasma as she is taken off the first Air Force rescue helicopter at West Yellowstone. She died Tuesday night in Deaconess Hospital, Bozeman. (Wide World)



Loading victims off Air Force helicopter at West Yellowstone.

(United Press International)



Settling quake victim in West Yellowstone Airport shed before flight to Bozeman.

(Salt Lake Tribune)



Nurses care for Mrs. Warren Steele, 37, of Billings and other quake injured campers in the interior of Johnson Flying Service DC-3 on flight from West Yellowstone to Bozeman.

(Wide World)

“As Mrs. Greene took us around, and gave the case histories, we saw what a resourceful job she and another nurse, Mrs. Fred Donegan of Vandalia, Ohio,

had done in the absence of drugs, medications, and even proper bandages. We helped with the dressings we'd brought, and the medicines for pain and shock.

“We were there an hour to an hour and a half. These injured needed hospital care, and there were no plans, as yet, to get them out. They couldn't travel by boat. So we got in our boat and went back to West Yellowstone to arrange for the injured to get from there to Bozeman and to the hospital when the helicopters did arrive.”

45



Slide victim Ray Painter, 46, of Odgen, Utah is carried by volunteers from helicopter to evacuation plane at West Yellowstone airstrip. (Wide World)



Reporters, photographers, and doctors all giving attention to quake victims, resting on bales of hay at West Yellowstone Airport hangar shed. (Salt Lake Tribune)

Shortly after noon, the first helicopter, a two-rotor silver Air Force H-21 from Hill Air Force Base, Utah, took its first load of four injured from the dam.

46

At West Yellowstone Airport, as arranged by Dr. Bayles, these injured—in sleeping bags—were immediately loaded onto the floor of a converted B18, which had brought cargo to “West,” and flown to Bozeman. There Dr. Bayles had organized a fleet of station wagons to rush them to the hospital for the care that was to save most of their lives.



Associated Press reporter Robert Moore interviews victim Warren Steele of Billings at West Yellowstone Airport.
(Wide World)

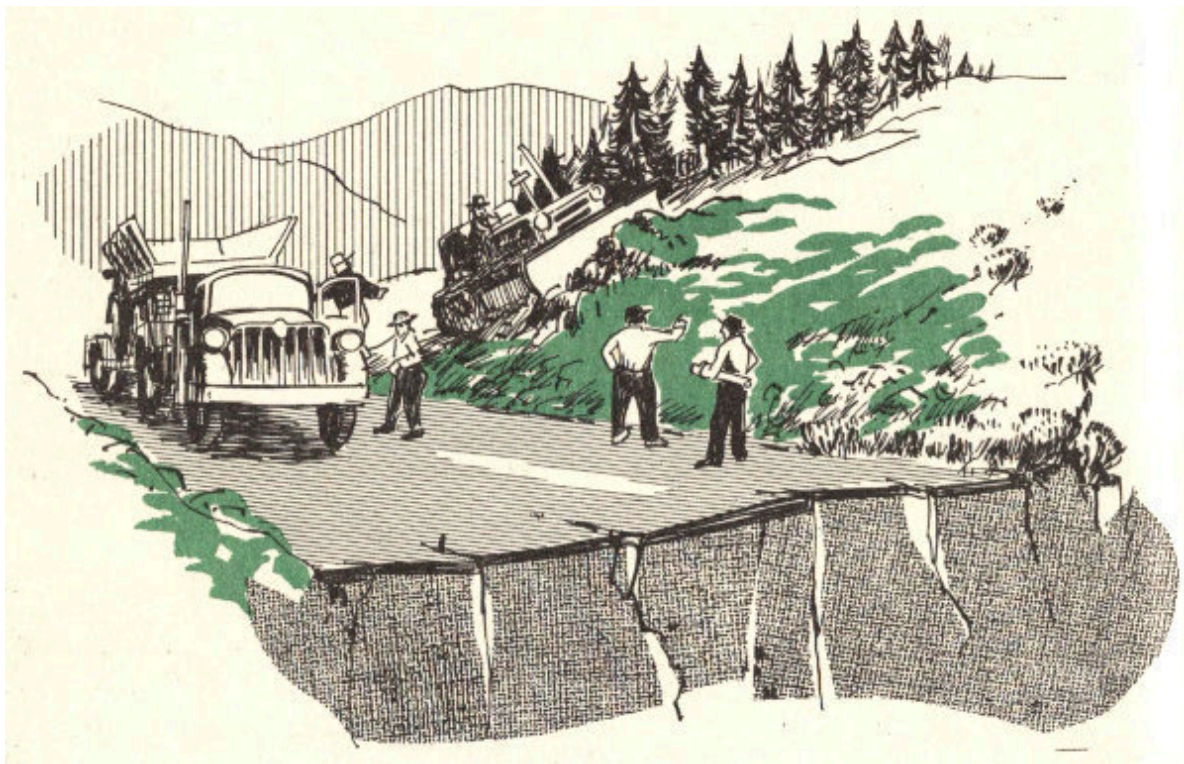


Dr. Bayles, first M.D. to reach trapped quake victims, watches as rescued are loaded off Johnson Flying Service DC-3 at Bozeman Airport.



Station wagons were pressed into service as ambulances, to take victims from Bozeman Airport to the hospital.
(Oliver Campbell, Manhattan, Mont.)

CD WRAPUP



By 3:45 o'clock the highway department was in full action. Major road repair help was on the way to get the roads open. George Barrett at the department's Bozeman HQ called Spike Naranche of the Naranche & Konda contracting outfit, which was building a big stretch of road in the Gallatin Valley, about forty miles north of West Yellowstone, and got their big-scale road-building equipment rolling toward West Yellowstone and the Hebgen Dam area. There was still no definite idea of the exact damage or the road blockage, but they'd begun to suspect major damage to the dam, the roads, or both. If the Highway Department couldn't use the equipment for road repairs, the Power Company could for dam repair.

Pilot Ralph Cooper took off in the Fish and Game Commission plane at 3:45 A. M. from Helena to reconnoiter the Madison area. Shortly

thereafter Quinnell and Alex Stephenson took off in the Highway Department's plane.

With daybreak came the first word on just what had happened. At 6 o'clock the planes reported (CA-1) as recorded in the Highway Department's log.

"Slide area 43 mi. so. of Ennis. White sign on the top of dam reading OK-SOS. Road has gone into the lake on the road side. Mountain has gone into lake on opposite side. Cracks 6 to 8 ft. across the road. Slide is estimated to be ½ mi. long and 300-500 ft. deep. Water rising fast. About 50 cars stranded in the area. Estimated 150-200 people. The only way out by helicopter."

Potter immediately called Johnson Flying Service, a pioneer regional flying outfit in Missoula, 200 miles from the slide, and ordered a helicopter for rescue work. He also asked for helicopter assistance from Malmstrom Air Force Base in Great Falls, Montana, 190 miles to the north. Malmstrom's rescue copter had blown a tire the day before, so they sent a jet to Salt Lake for a new one. Potter hollered for helicopters on the National Alert Warning System hot line.

"How many do you need?" he was asked.

"All you can get," he answered.

In response, everything, flying amphibians, transports, in addition to helicopters, started moving toward the quake area—from the 41st Air Rescue Squadron, Hamilton Air Force Base in California, the 2849th Air Base Wing Rescue, Hill Air Force Base, Utah, the 3638th Flying Training Squadron, Stead AFB, Nevada, and the 4061st Support Group, Malmstrom AFB, Montana.

The Forest Service began moving in its well-organized rescue organization that morning, under the direction of Harvey Robe. Eight of the FS's elite smokejumpers, trained in first aid, jumped in the canyon at 10:30 o'clock, with rescue equipment under the leadership of Al Hammond.

"When we made our parachute landings," Hammond remembers, "The folks we came to rescue asked us, solicitously, if we were OK."

The rescue of the people trapped in the canyon—it turned out that there were close to 300—proceeded smoothly. The Osts, Fredericks, and Smiths, 50 all ambulatory, if shoeless were helicoptered out to the highway on the Ennis side of the slide, and taken in highway patrol cars to the hospital or to the dormitory improvised in the high school gym. The injured who'd been gathered at the Hebgen Dam end of the canyon were helicoptered out to West, and flown to the hospital in Bozeman.



The slide that blocked Madison Canyon, dammed the river, and brought terror and tragedy to those at Rock Creek Campground as viewed from the valley side. (U. S. Forest Service)

Working continuously through the day, without provisions for meals, etc., the road repair crews “barbered” a shoo-fly substitute exit road along the steep mountainside parallel to the shore where the road had collapsed into the lake. By 6:00 P. M. they’d completed a passable road. The State Highway Patrol registered the cars as they exited from their entrapment in the Madison Canyon. When all the unencumbered cars had passed through, the bulldozers helped pull those with trailers over the most difficult portions of the substitute road. That night the refugees were welcomed to food and beds in the Montana State College gym in Bozeman. 51

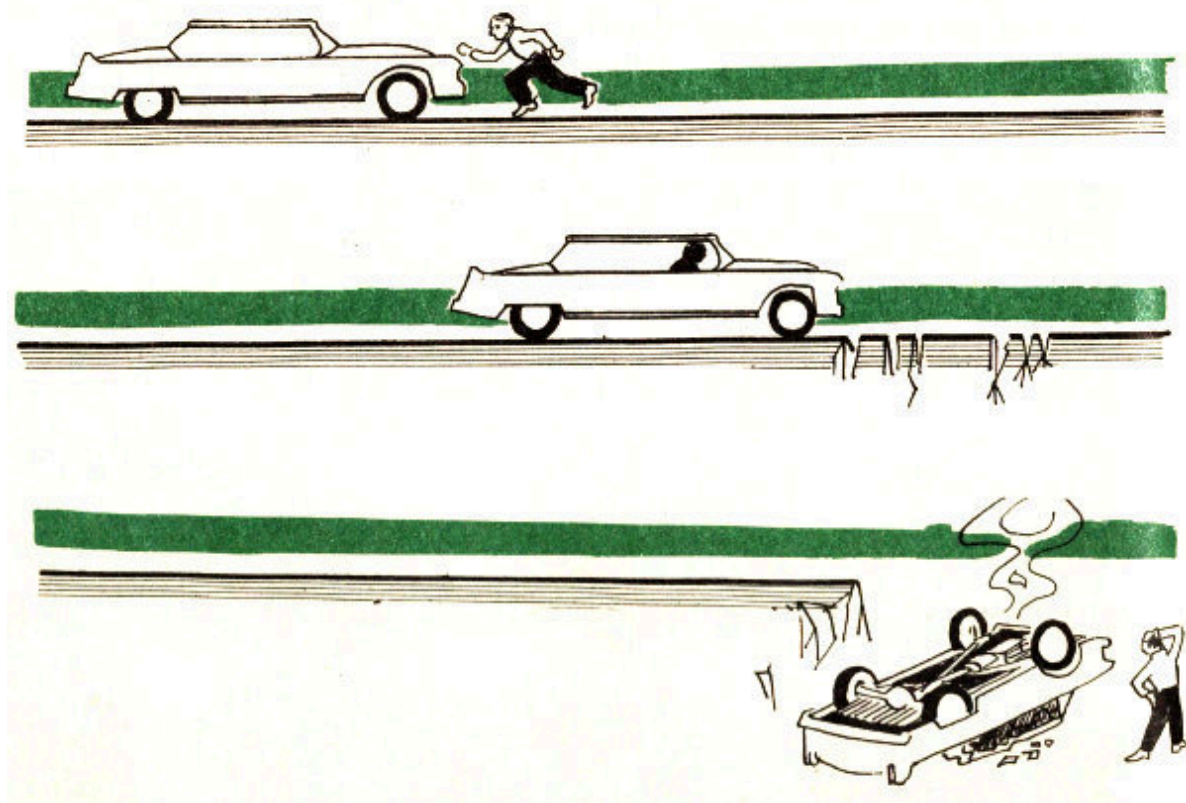
Within eighteen hours after the initial shock, the last of those trapped by the earthquake in the difficult-to-reach Madison Canyon were on their way to safety. The wounded had been rescued hours before. As George Sime, information guy for the Highway Department and for CD, said,

“That day anyone would have been proud to be a member of the Highway Department.”

The whole operation ran smoothly—it was a tremendous example of government service in the finest tradition—a demonstration of agencies working together to do an important job.



Where the mountain fell—as viewed from helicopter approximately over the site of the buried area right next to Rock Creek Campground. (Montana Power Company)



Nobody held back. They put in all the personnel, and spent all the money needed to get it done.

“When we knew lives were at stake,” Forest Service Region 1 Chief Charles Tebbe said, “We didn’t worry about the cost or what appropriation it would come from. We just went ahead and did the job.” Quinnell, head of the Montana Highway Department, took the same attitude.

It wasn’t until three days after the quake that anyone mentioned the fact that no one, including Potter, actually had authority for much of the work they’d done. It belonged to the sheriffs of the counties involved. By this time the emergency job was practically done. All that remained was to figure up the damage.



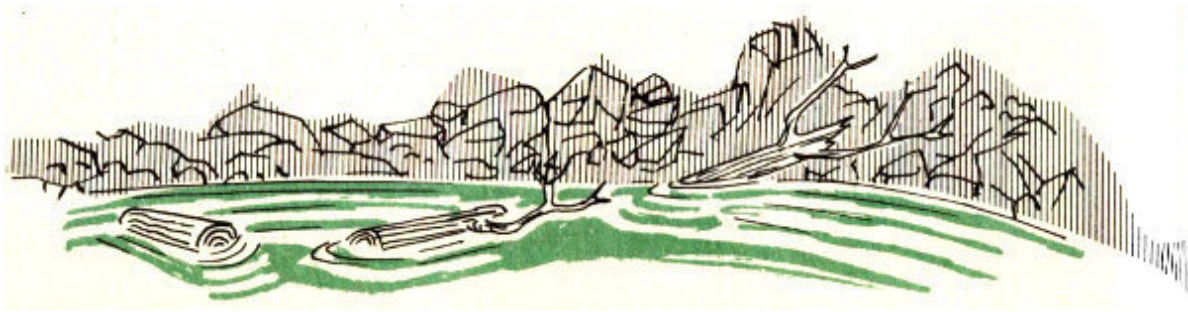
Two of the road drop-offs, and the shoo-fly, or improvised escape road built the day after the quake trapped 300 vacationers in Hebgen Canyon. (Christopherson)



(Bottom Forest Service)

UNTRAPPED

All through the night, the Osts, the Fredericks, and the others trapped above the slide shuddered with each new quake, and then listened for the repeated thunderous crashings of the avalanches which echoed loudly against the canyon walls. Every 15 to 20 minutes all that morning there would be another shock.



They were thankful that their families were complete. Fredericks, nearly exhausted from his work in helping rescue those trapped by the rapidly advancing water above the slide, tried to sleep, but the excitement and uncertainty kept the whole group awake. At dawn, which came at about five, the first of the many small planes flew over the canyon. The light gave the group a clear view of the opposite side of the canyon, and they could see how the mountain had turned loose, crashing down onto the canyon floor, surging up the other side of the canyon to a height two-thirds of the height of its original location, and then shooting both up and down the canyon. They could now see the mud, debris and the accumulating water which had covered their cars and camp.

In the early light they used merthiolate and dressings from a first-aid kit to treat the worst of the previous night's injuries. The two dozen eggs, somehow rescued intact in their flight up the canyon side, fried with canned potatoes and served on bread, plus coffee made a heartening

breakfast. The Smiths, who'd fled the Beaver Creek Campground at the time of the quake, joined them, making a total of 21 in the group.

A small, orange and silver plane swooped low, circled, and waving its wings, flew east toward the dam. They took heart in the fact that they'd been discovered.



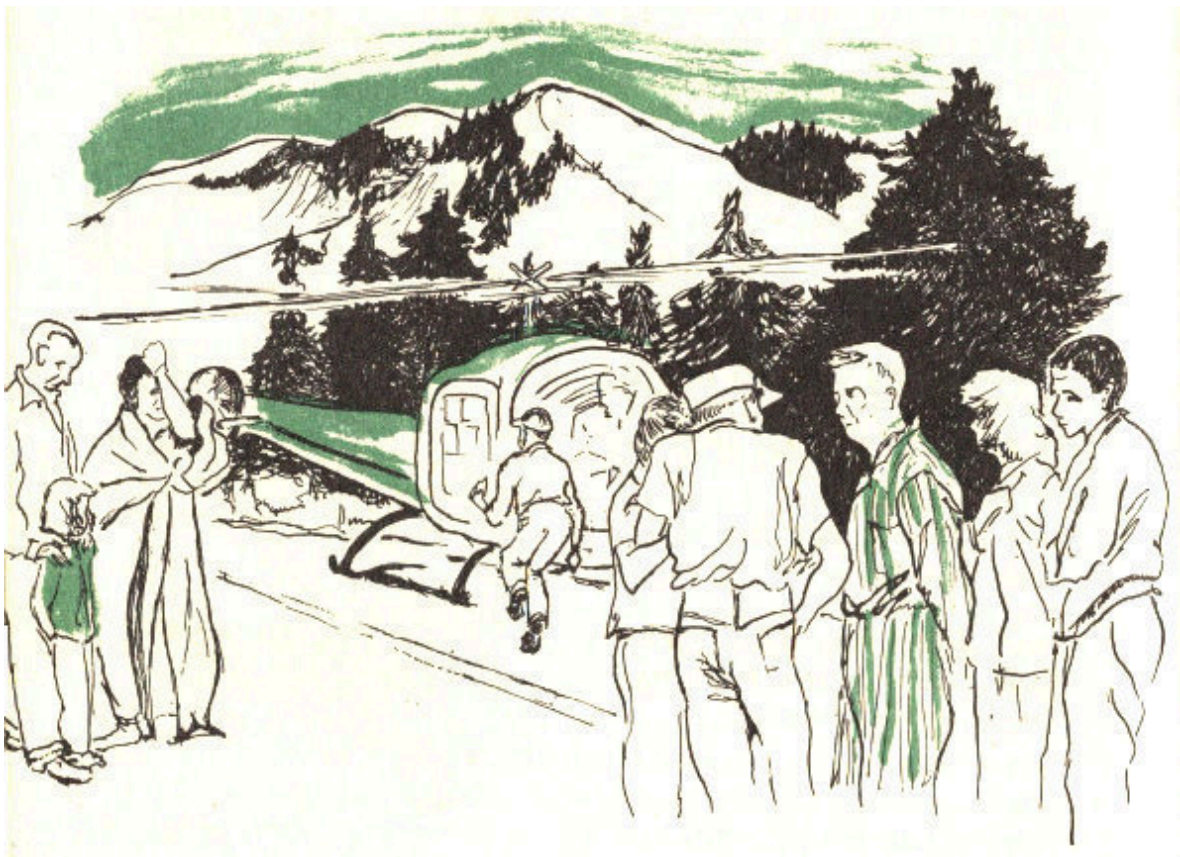
Half an hour later, the plane flew over again, very low, dropping an orange streamer fastened to an envelope. The envelope was torn open by a branch,

and the message floated down by itself. With fresh horror they read it. It said, "Fire down by river bridge on ridge top. Get going."

It was signed simply, "Ost."

Hurriedly they looked around for smoke. Seeing none, frightened, trapped in a strange, wild country, with all nature seeming to turn against them, they knew not where to turn.

56



In an effort to find out about the fire, Ost borrowed the hip boots a woman had given Fredericks and started off in the direction of the slide. The plane circled over him and wagged its wings, an action he interpreted to mean that he was going in the right direction. He continued, climbing the muddy lower end of the slide, the rubble, the great cube-shaped boulders, big as cars, all mixed in with trees, some stripped bare, others still complete with all their branches.

On the slide he met two men walking in from the outside. They told Ost that the river bridge was 15 miles upstream, past the dam, and advised him to keep the group where it was until helicopter help came.

The plane message about the fire was still a mystery. It remained so for several months, until Ost finally got it explained. The message had been one of several dropped from a plane by a Forest Service guy, Otto H. Ost, 57 in 1957 to instruct a ground crew to proceed to a fire a couple of hundred miles from the Madison Canyon. The streamer, Otto Ost figured, had been found, returned, and sent out without removing the two-year-old message. The note from Ost to Ost was a powerful coincidence.

“Doesn’t it strike you as almost planned?” Rev. Ost said when he got the explanation.

At 11:30 A. M. the two Forest Service smokejumpers, part of a group of eight who’d jumped farther up the canyon, hiked in. They had first aid equipment and food. They reassured the group that a helicopter was on its way to rescue them.

Shortly after noon, the Johnson Flying Service helicopter arrived, landing precariously on the canyon-side slope. Mrs. Ost and George Whitmore, the Fredericks nephew, both with eye injuries, were the first two taken out. The helicopter ferried them over the slide to a point on the highway where Highway Patrol cars sped them—at 80 miles an hour where rocks hadn’t made the road hazardous—to Ennis, to medical care, comfort, and safety.

By the time the helicopter had taken seventeen of the group over, the turbulence of an oncoming storm made the air so treacherous that the four remaining men walked up the canyon, were driven to a safer landing point at the upper end of the canyon. They saw the cracks and damage at Hebgen Dam. The helicopter picked them up, and they joined their families in Ennis.

“We’d lost our money, our cars, our clothes,” Mrs. Fredericks said. “The Red Cross didn’t ask us any questions about whether we had any money or not. They just helped. They sent us to stores and got us all two complete outfits. They told us to make any calls home—to our relatives—that we wanted to. And they’re flying us home.

“We’re certainly going to be ardent Red Cross workers from now on!”

That night they stayed—dormitory style in the Ennis High School Gym—as Mrs. Fredericks put it, “An anvil chorus, each snoring in his own language.”

Conditioned by the quake of the night before, when the town siren blasted off at 9:00 P. M. the quake victims jumped out of bed in alarm and 58 hastily dressed. Even after the siren was explained as the regular nightly curfew signal, Mrs. Fredericks slept the rest of the night in her clothes.

“I’m damned if I’m going to be caught in my pajamas again,” she said.

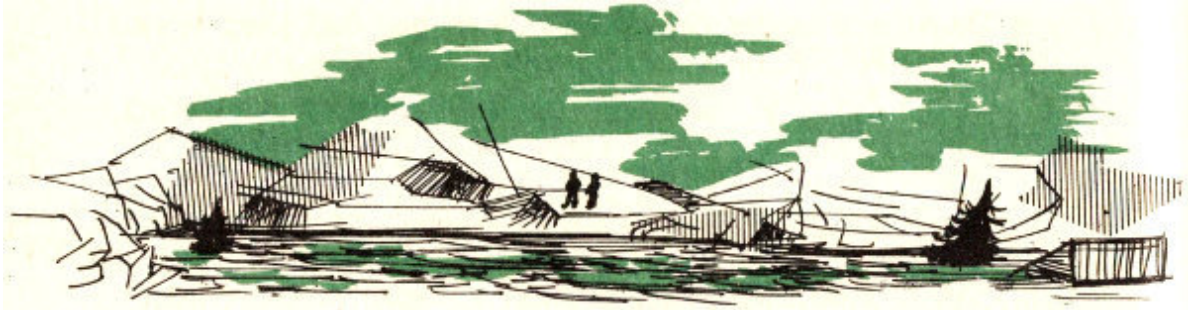
The Osts moved up to the Shermont Motel in Sheridan, where Mrs. Ost recuperated from her face and eye injury. Miraculously, though the whole side of her face was massively bruised, no bones were broken. That Saturday they were guests of the Madison County Fair at Twin Bridges. On Sunday the Red Cross flew them from Butte back to their homes in New York.

The Fredericks moved to the Finlen Hotel in Butte while George Whitmore had treatment for his more serious eye injury.

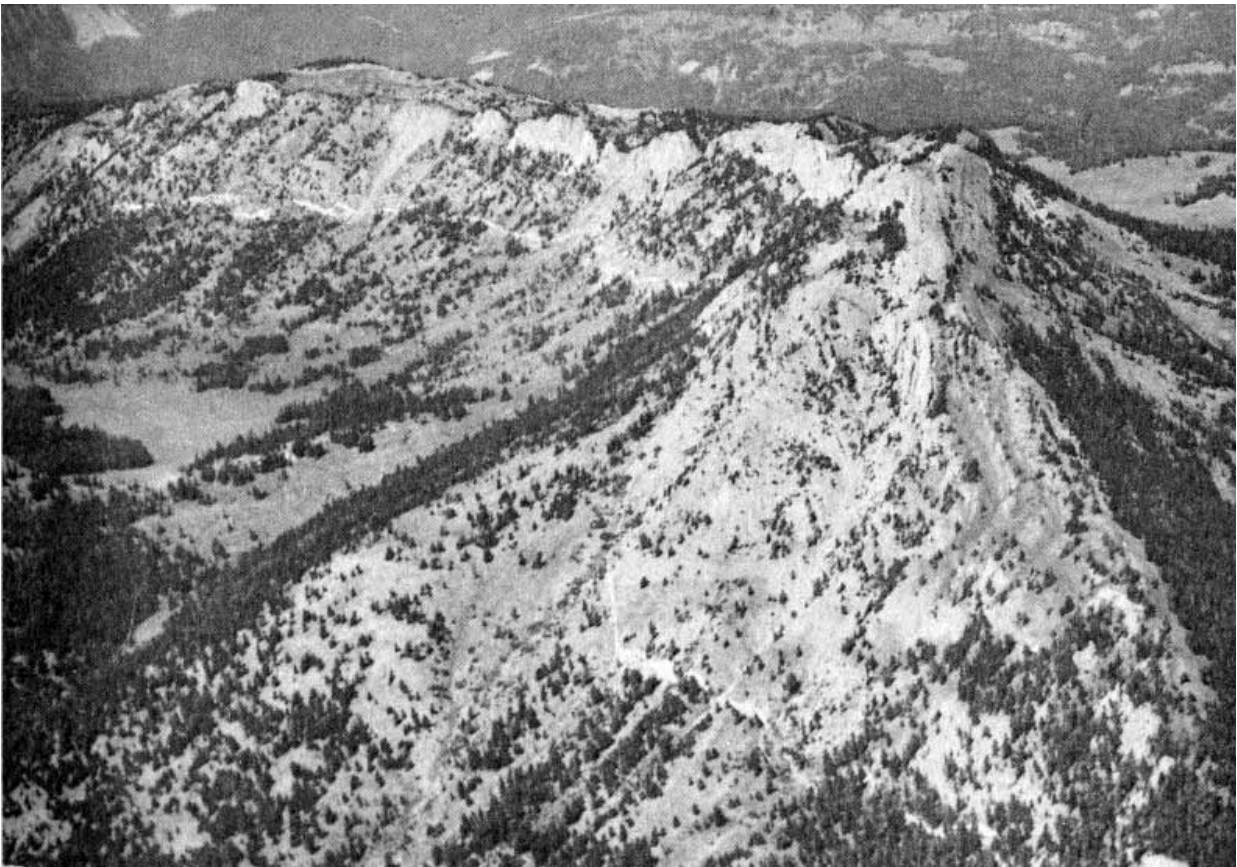
“Everyone was so wonderful. A bellhop drove us all around, showing us this exciting town. The people at the hotel took up a collection and gave us some money. You couldn’t have better people.”

The Fredericks flew back to Elyria that same Sunday, leaving George Whitmore in the hospital for further treatment.

“The irony of it all,” Mrs. Fredericks said, “is that we still didn’t get to see Yellowstone Park.”



59



The main, Red Canyon scarp runs for 7 miles along Kirkwood Ridge, parallel and about three miles above the north shore of Hebgen Lake. (Montana Highway Commission)



MYSTERY—WHO GOT IT?

With the primary emergency—rescuing the injured and freeing the trapped—contained, there still remained the perplexing problem of trying to figure out, just how many were still buried under the 80-million-ton rock slide.

Early guesses put the figure in the hundreds.

The infeasibility of moving 43 million cubic yards of collapsed mountain in quest of this gruesome total was almost immediately apparent.

Aerial photos taken that next morning showed that the slide hadn't covered the improved portion of Rock Creek Campground—the section with five formally-laid-out campsites, each with parking spot, fireplace, picnic tables, etc. But this didn't help much toward an estimate of the total buried, because of the informal fashion in which both trailer and tent campers would set up for the night anywhere they could find a level spot.



Water from normal flow of the Madison beginning to flood the canyon, and Rock Creek Campground, which was located under the water at the right. Note the streaks showing the flow patterns of the slide.
(Montana Highway Commission)

Fred Brauer of the Forest Service Fire Control Division in Region 1, HQ in Missoula spent quite a bit of time talking to survivors and others who might help to establish a probable total. 61

Rev. Ost, who'd been camped in Rock Creek Campground, said that he had counted 21 trailers between the mouth of the Madison Canyon and the spot he camped. The undertaker in Ennis, Charles Raper, put his estimate at 100 to 160.

Brauer found that Marshal George Hibert of the town of Ennis had been on a fishing trip in the vicinity of the slide area on Monday, August 17, and for some inexplicable reason decided to cut the trip short, leaving the area at 9:30 o'clock that night, about two hours before the quake. He guessed, from his observations, that there could be 100 people under the slide.

Guy Hanson, a West Yellowstone 5th and 6th grade school teacher who was working that summer as a Fire Prevention Guard for the Forest Service, had periodically checked the Madison Valley campgrounds. In one survey, shortly before the quake, he'd found five tents, eight trailers, and 42 people in the Rock Creek Campground. In the adjacent, unimproved area, he'd counted another 25 people. At noon the Monday of the quake, he'd helped police the improved area and found six trailers parked there at midday. He didn't check the unimproved area.

From his familiarity with the campground, Hanson felt that 40 campers were probably trapped under the slide. Brauer chilled as Hanson told him that the occupancy could be higher if there were groups, such as Boy Scouts, camped there. He described how often he'd found scout encampments at Rock Creek, and how one scoutmaster had told him that this was a favorite camping objective for many Utah scout troops.

While Brauer was trying for an answer as to how many, others were working on the question of just who was buried under the slide. Mrs. T. Mark Stowe of Sandy, Utah, was considered a probable from the first. Her husband was washed out from under the lower end of the slide, and it was logical to deduce that she'd been caught under the rock mass.



Whose gear? This collection of clothes and creel found below the slide, may be a clue to one of the victims still buried under huge pile of rock and debris. (Lewis—Montana Standard)

Volunteers from Ennis, Butte, Virginia City and elsewhere walked over the rugged slide for days searching for any kind of clue which might help. They turned up everything from fishing creels, camp equipment, and souvenir pillows to kids' shoes. One slide-walker found an exposed roll of film which was immediately heralded as a hot lead. But when developed, the film turned out to have been ruined by the water. Their findings were kept in a county warehouse in Ennis. Much of it was claimed by those who'd escaped. The debris, so painstakingly gathered, helped little in the search for identity.

The quest evolved into the painful job of waiting, keeping lists, sifting names.

From the moment the quake occurred and the fact that there were casualties became known, phone calls, telegrams, and letters began surging in from all over the U. S. wanting to know about friends and relatives who might be in the area or among the victims.

63



Volunteer slide workers searched the huge 80-million-ton pile of rock for victims, and clues to identity of the buried. (Lewis—Montana Standard)

“The Disaster Service of the American Red Cross did a tremendous job through their teletype and telephone by taking over these inquiries and sending back information through the Home Service chapters,” Don Skerritt, Sheriff of Gallatin County, said.

One of the leads was a spaniel discovered wandering in the slide area the day after the quake. The animal wore a Salt Lake dog license tag. This seemed like a certain clue to the identity of some of the victims.

In response to Skerritt's teletype inquiry, Salt Lake police found that the dog supposed to be wearing the tag had been killed months before. Someone had hung the collar in a gas station. Subsequently this collar was put on another dog. Further probing developed that the dog in the quake area belonged to the Ray Painters of Ogden, Utah. Mrs. Painter, one of the casualties, died in the Bozeman hospital a couple of days after the quake.

64



In Ennis County Warehouse, Red Cross Area Disaster Director Ralph Carlson, who flew to the quake scene from San Francisco, looks puzzled as he surveys part of the truckload of material recovered by slidewalkers searching for clues to identity of those buried under slide.

(Christopherson)

Skerritt, who's like a quieter, shorter version of Kefauver, worked all that week and for months afterwards on the identity problem. During the first weeks, Red Cross volunteers and personnel worked around the clock to answer the flood of inquiries. There were some 3,000 of them. They felt fortunate that no scout troops turned up missing.

These queries, they painstakingly sifted, sorted, and winnowed down. With tireless persistence, they kept at it, writing to the source of each of the thousands of inquiries to find out if the missing had turned up. New inquiries kept coming in—and still do, asking about people that just plain haven't been heard from, and their relatives, or friends have thought of the slide as a possible explanation.

Through tangible tie-ins, like postcards, letters, the use of credit cards in the area just before the quake, phone calls from the area, they finally got down to a list of those highly probable as slide victims whose bodies will never be uncovered.



Gruesome reminder? Days after the quake, these children's shoes and clothing still lay in the dried-up streambed below the massive slide. (Christopherson)

Take the case of Roger Provost, an official at the California State Prison at Soledad, California. He had been in touch with his office up to the date of the quake. He was a methodical type. Upon leaving California, he left a planned vacation itinerary stating that the family was to proceed from Yellowstone down the Madison River (August 18, 19, 20 and 21) and to Bozeman, etc. Several cards to friends and relatives postmarked Aug. 16 at West Yellowstone, Montana, stated that the family was at a trailer campsite "on the Madison River, about 30 miles from Yellowstone."

Another family, Robert J. Williams and wife, Coy, children Michael, 7; Steven, 11; and Christy, 3, of Idaho Falls, had told relatives they planned on fishing the Madison River. They registered as visitors at the Museum in Virginia City on August 17, 1959, and weren't heard of again.

66



Dr. Merle Edgerton and his wife, Edna, in their car and trailer, were travelling with Harmon Woods and his wife, Edna, who also brought their car and trailer. Dr. Edgerton kept in daily contact with his hospital in Coalinga, California, up to the time of the slide. He was last heard from on August 15, giving the families' location as "on the Madison River, outside Ennis, 35 miles from Yellowstone."

The complete list of those who on such evidence are considered buried under the monumental Madison Canyon slide totalled 19. They are:

Sidney D. Ballard, wife, and son of Nelson, B. C.

Bernie L. Boynton and wife, Inez, of Billings, Montana.

Dr. Merle Edgerton and wife, Edna, of Coalinga, California.

Roger Provost, and wife, Elizabeth, and sons, Richard 16, and David, 1½, of Soledad, California.

Mrs. Thomas Mark Stowe of Sandy, Utah.

Robert J. Williams, and wife, Coy, and children, Steven, 11; Michael, 7, and Christy, 3 of Idaho Falls, Idaho.

Harmon Woods and wife, Edna, of Coalinga, Calif.

The other quake casualties include Mr. Purley Bennett and children, Tommy, Carole and Susan of Coeur d'Alene, Idaho, who, along with Thomas Stowe of Sandy, Utah, were found below the slide. Mr. and Mrs. E. H. Stryker of San Mateo, California, were killed by the boulders at the Cliff Lake Campground. Mrs. Myrtle Painter of Ogden, Utah, and Mrs. Margaret Holmes of Billings, Montana, died of quake injuries in the Bozeman Hospital.

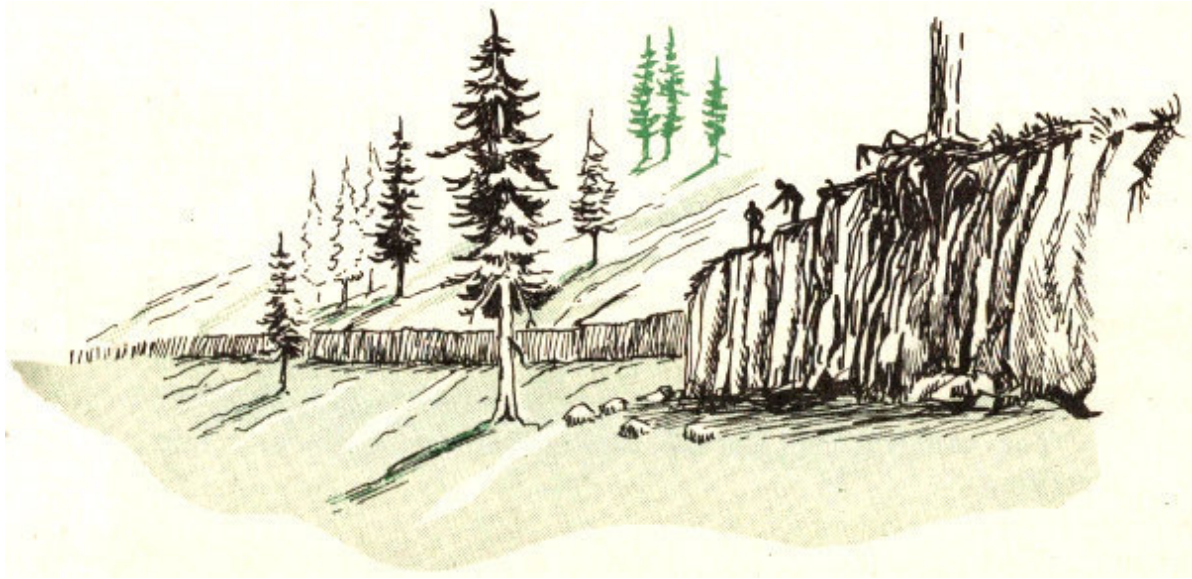
The final Montana-Yellowstone Earthquake death toll stands at 28.

CD AND NATURAL DISASTERS

Natural disasters, like the Montana-Yellowstone Earthquake, are perhaps the best test of our Civil Defense readiness. Until the quake, CD Director Hugh Potter wasn't at all sure that he had an outfit at all. Operating on a short budget of \$21,000 a year, with all of the third biggest, sprawled-out state to organize, he'd set up, at least on paper, state-wide and county CD groups. He'd compiled an exhaustive inventory of state facilities, resources, etc., complete to such minutiae as the number of aspirins in the state (1,657,000 5-gr. tablets), and the amount of meat Montana's abundant wildlife represented (58,517,725 lbs., including 1,000 bison, 415 grizzly bears).

The alerts he'd organized weren't notable successes, and he'd caught some hell from the higher-ups for not being current on the state-wide alerts which are supposed to be held at least once a year.

"Our people just aren't too enthusiastic about practice alerts," Potter says. "Frankly, they feel it's a waste of time. They're busy. They don't want to play war. A guy will say, 'I want to go fishing, or put up a hay crop, or something.'



“But let a real emergency happen and they’re right there.



Real shook—the interior at the colorful Virginia City Courthouse the morning after the quake.

(Lewis—Montana Standard)

“During that first day Ed Cottingham and I were busy pulling triggers. I realized then that the most important thing I’d done during my seven years as CD director was getting around and getting to know who to phone—the people you can count on to get something done in an emergency. You can get the heck of a lot done if you know the right guy to call.

“There isn’t a CD department that didn’t check in right away to find out if they were needed.

“We’re especially lucky to have the U. S. Forest Service (the big slide happened in the Beaverhead National Forest) in our area. Their experience and constantly organized readiness to meet the threat of forest fires right now makes them an ideal outfit for any emergency. Forest Service

firefighting squads, transport, equipment, and information about the area are all set up to move in a matter of minutes. They're most adaptable to the kind of crisis the earthquake threw at us.

"You can tell the Forest Service your problem and quit worrying.

"Another important outfit is the Montana Forestry Department, which is set up to administer and protect the state's forests. Its boss, Gareth Moon, is head of the CD's Rural Firefighting Section. 69

"We have a good, mobile law enforcement outfit in the Montana Highway Patrol. The Montana Fish and Game Dept. men, in emergency, serve as an excellent backwoods force.

"Frank Wiley, Montana Dept. of Aeronautics director and one of the real pioneer pilots who can still fly anything from a jet to a Jenny, took over our flying problems."

At 8:45 A. M., as part of a CD emergency plan called "Operation Bulldozer," set up by the Associated General Contractors, Jack Marlowe, secretary of the Montana Contractors' Assn., had completed a list and location of all heavy construction equipment in the area and reported that all contractors were on standby in case they were needed.

The State Dept. of Health was on the ball, too. They were moving in personnel to test water in Ennis, West Yellowstone and throughout the quake area by 9:00 the morning after the quake.

At 9:15 word came in that the Red Cross was flying in emergency personnel from the west coast.

Potter was thrilled by the offers of help that kept CD HQ phones busy. General Keith R. Barney of the Army Corps of Engineers called offering any help needed. The governors of Idaho and Wyoming and three Canadian provinces asked if there was anything they could do. Idaho's highway patrol actually came up and helped keep things under control in the West Yellowstone area.



Several search and rescue outfits called, offering aid. A combined Army, Navy and Marine Corps Reserve unit from Butte gathered their medical equipment and ambulances and sped to the Ennis side of the slide as a voluntary, unpaid action.

There were offers from the crack mine rescue teams from the famous 70 Anaconda Company mines in Butte. When a call went out on the regular radio for housing for the Ennis evacuees, several hundred accommodations were phoned in to a local Butte station. Another abortive suggestion that men on horseback might be needed to search some of the impassable back country brought over a hundred volunteers in less than an hour. A Bozeman station was overwhelmed with offers in response to an announcement that station wagons would be needed at the airport to ambulance the wounded to the hospital.



Nurses, doctors, National Guardsmen, skindivers—they all called in wanting to help.

At Great Falls, where the Montana Red Cross Blood Bank was holding a regularly scheduled drawing, when word came that they were flying blood to

Bozeman to help the victims, so many volunteers showed up that the total exceeded 450 pints, and at closing time 150 donors were still in line.

“We had special problems—distance from any sizeable town was one. I’d hate to think of the casualties if the quake hit in a really populated area,” Potter said.

“The mountains, which obstruct short-wave signals and set up all sorts of radio blind spots, made it difficult to get any sort of ground radio communication going. It was impossible from the ground, to signal to Ennis or to Hebgen Dam from West Yellowstone. The radio amateurs did a tremendous job of helping those first few hours—they set up a standby network and kept it clear for emergency messages. One ham, Father Francis A. Peterson, (W7RKI), from St. Anthony, Idaho, one of the first to report the quake, loaded up his gear, drove to West Yellowstone, and by 7:45 A. M. the morning after the quake had set up radio control at the otherwise 71 radioless West Yellowstone Airport. Another ham, Harold L. Beddor, (W7JPD), of Dillon, Montana, handled emergency communications from Dillon the day after the quake, then flew his equipment to the West Yellowstone Airport to help out too.

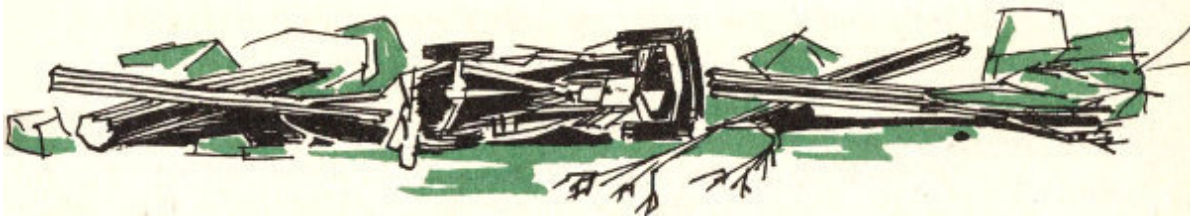
“The problem was complicated by the multiplicity of wave lengths on which the various civilian and military agencies were operating. We discovered that the high frequency bands the Civil Defense uses (150 megacycles) are useless in mountain country, especially in the day time. Lower frequencies, 34.82 and 46.86 megacycles did get through.

“All that first day or so, we relied on the Highway Department plane, which was radio-equipped, to get messages across the quake area. It stayed in the air all day.

“The mountain altitude (at the dam the elevation is 6,550 ft.), presented aviation difficulties, too. Smaller helicopters couldn’t make it, and some of the bigger jobs were tricky to fly in the less dense mountain air.

“We had difficulty with aerial sightseers. In spite of our announcement that the fields at West Yellowstone and Ennis were closed to all but emergency aircraft, planes flew in from all over. Charter pilots flew in from as far away

as Arizona, and did a brisk business in flying the curious over the quake area at \$6 a head. Including the Air Force ships, during the first few days the West Yellowstone Airport was as busy as Chicago's Midway Field, with planes taking off and landing at the rate of one a minute. With all the traffic over the slide area, it was a miracle that we got through that first week without a crash.



“But as a result of the quake we know that any area which has this kind of emergency will make out OK with the wonderful spirit of people, helping and wanting to help.”

AFTERMATH

Among the cataclysms of nature, none is more terrifying than an earthquake, and huge slides, like the one triggered by the Madison Canyon earthquake, are perhaps the most dramatic type of geologic change. In one sudden, spectacular moment, changes take place that make us think of the tremendous energy released by atomic fission, the earth's mass moves in a volume that rocks the imagination, and its effect on the people who are near, or in the path of nature's huge, impulsive-seeming change helps us to realize how infinitesimal we are before the forces and laws of nature.

In 1903, a 40 million cubic yard rock slide crashed down from the crest of Turtle Mountain onto the coal-mining town of Frank, Alberta, killing 70 people.

But the consequences of such huge slides aren't completed when the cliff toppling ceases. Take the case of the famous June 23, 1925, Gros Ventre slide in northwestern Wyoming, 40 miles south of Yellowstone Park. An estimated 50 million cubic yards of rock and debris plunged down the steep canyon wall, shot across the valley floor, and rushed some 350 feet up the opposite wall of the canyon before it settled back, like water sloshing in a huge bowl.

Nobody was killed when this slide choked the Gros Ventre River. It covered parts of two ranches and buried six head of cattle. But two years later, in May, 1927, the water dammed by the slide pushed out a big section of the slide and the sudden wave of water and debris washed away the town of Kelly, Wyoming, killing seven people.

This kind of possibility was in the mind of Army Corps of Engineers Missouri River Chief Keith R. Barney as he and Lt. Col. Walter W. Holgreffe of the Corps district offices at Garrison Dam in North Dakota, discussed the Montana earthquake's action. The slide represented a double threat to people in the Madison Valley below.

As an immediate effect of the slide, the water flowing over Hebgen Dam was stopped by the slide. The formation of a lake behind the slide began the moment of the slide. When it filled, this 240-foot deep impoundment, called Earthquake or “Quake” Lake, would exert an enormous pressure on the slide. If the slide was composed of unstable material, its collapse could, in a repeat of the Gros Ventre tragedy, bring death and destruction to the valley towns of Ennis, Three Forks, and Trident below.

73



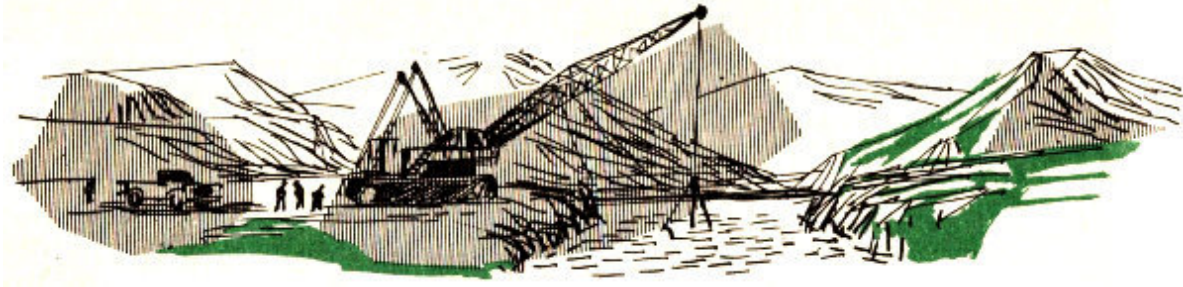
The Hebgen slide on September 10, when Quake Lake first began to flow through the 14-ft. deep channel cut by the Army Corps of Engineers. The immense erosion that followed prompted another 50-ft. channel cut on the slide. The lower picture shows the completion of this project on October 17, 1959.
(U. S. Army Corps of Engineers)



The second and greater threat was the discovery that when Quake Lake 74 filled, its impounded water would lap at the foundations of Hebgen Dam, and quite possibly undermine it, releasing a volume of water seven times that of the earthquake-caused reservoirs, which could also sweep part of the slide along in its mad rush.

Like the threat of a time bomb, the rising level of Quake Lake, and the increasing pressure of the water against the slide, augmented by rumor, kept the downstream towns in constant anxiety.

The Army Corps of Engineers rushed into emergency action. They flew in a 50-man-staff, and set up headquarters in the Stagecoach Inn at West Yellowstone.



The mines in Butte were on strike, and huge earth moving equipment from the open pit operations, along with rigs from other contractors, worked around the clock to cut a 250-foot wide and 14-foot deep channel across the mile and a half long slide and armor it with rock so the water couldn't cause sudden erosion of the rest of the slide before the water topped the huge natural dam.

On September 10, water licked over the new spillway, running into the river bed just below, which had been dry since the quake. To the Corps's great surprise, severe erosion tore the downstream face of the slide. To remedy this, they launched another crash program to cut a 50-foot deep channel across the top of the slide. It was completed October 29.

The Corps's operations on the slide took a total of close to \$1,700,000 of funds set up for such emergencies. As a result, the towns below the slide are safe from the flood threat the slide might represent.

LIVING GEOLOGY

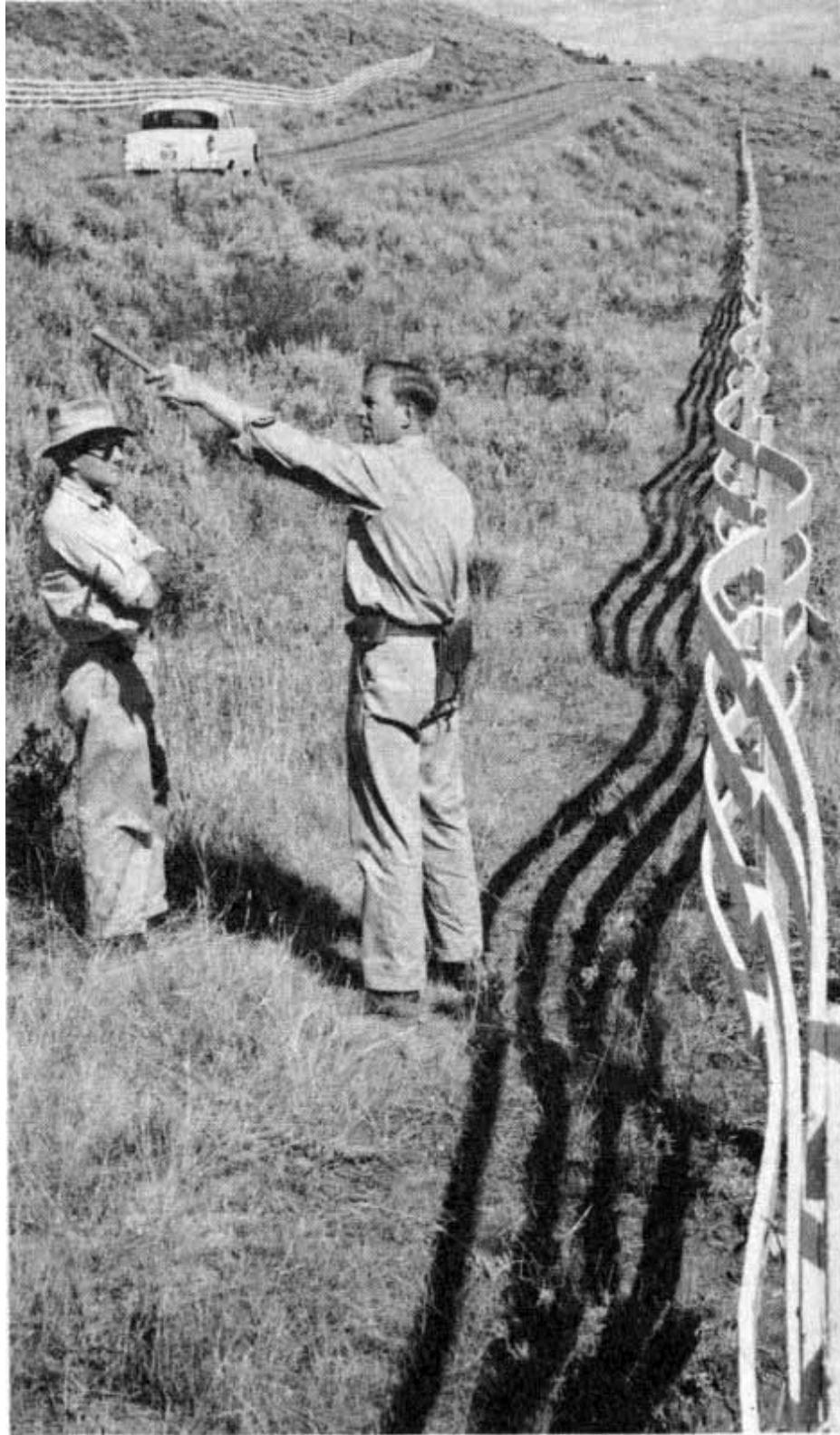
Just how common are mass earth shakeups like the Montana-Yellowstone Earthquake, anyhow?

Geologists tell us they're frequent, with a dozen or more major quakes, and thousands of minor tremors happening each year.

Earthquakes are the natural outcome of the fact that the earth, while seeming substantial and changeless, is constantly, if most gradually, in the process of change. Mountains are thrust up. Glaciers carve them down. Volcanoes pour out their molten rock. Rivers and floods scour their erosive paths. Sediments slide and settle.

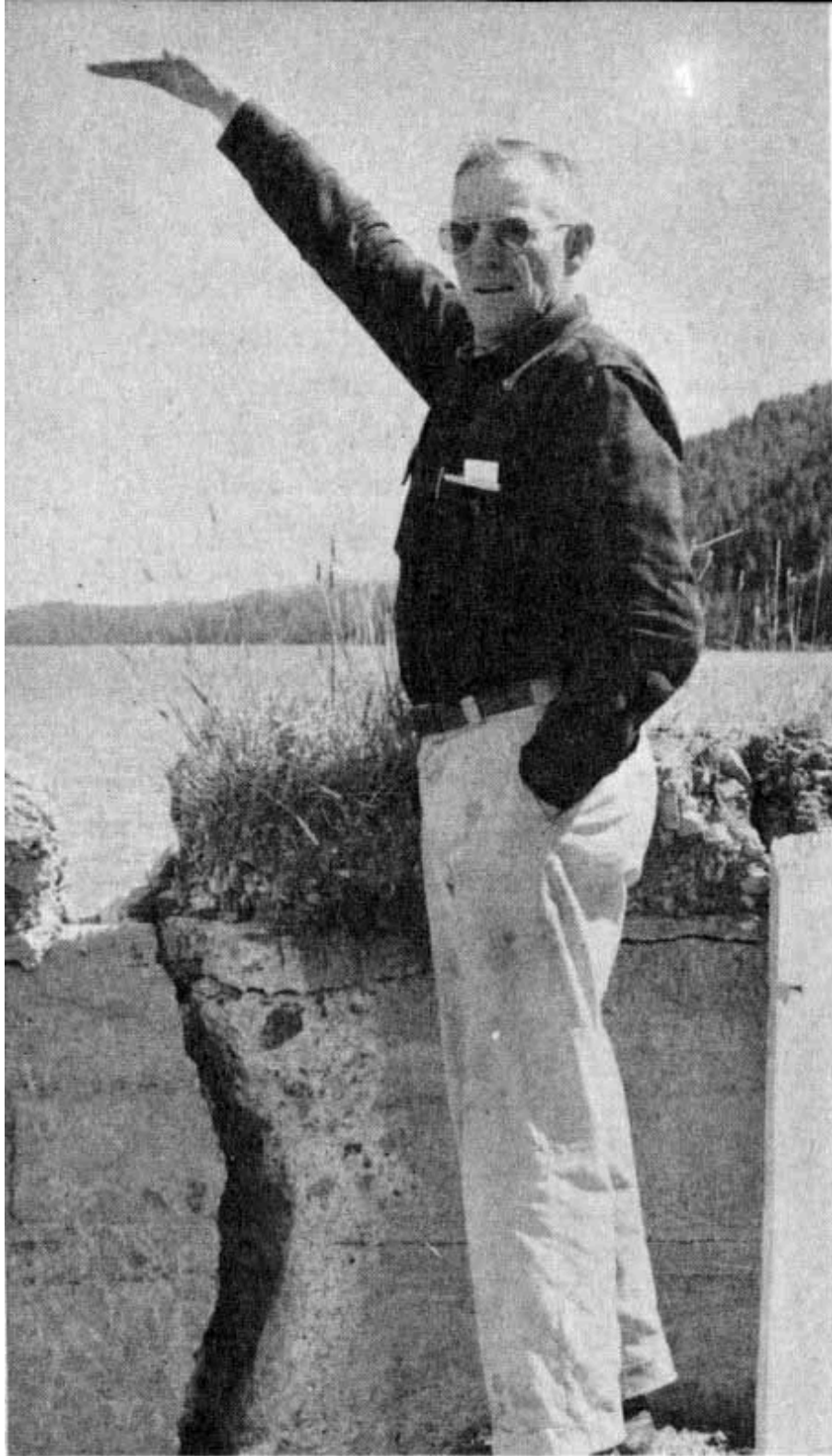
The enormous masses which great internal earth forces have raised up to mountain height, create counter stresses. These forces build up for years, sometimes for centuries or longer.

Eventually something has to give. When this happens on a grand, or spectacular scale, we call it an earthquake.



Stress, caused by the quake, resulting from slump of surficial material caused the attractive curving of this fence along Highway 287 east of Hebgen Dam. (U. S. Geological Survey)

Whether you're a connoisseur, expert, or not, the spectacular 1959 Montana-Yellowstone Earthquake was a beaut. Geologists call it a "textbook earthquake," because it included nearly all of the classic actions, or results which quakes are likely to cause.



Damkeeper George Hungerford points out how much the tidal waves, or seiches, overtopped Hebgen Dam at the time of the quake. Hungerford is standing on earthfill washed down from dam-top level by the gigantic waves. The quake also cracked the dam's concrete core as shown.

(Montana Power Co.)

It ranked right along with San Francisco's 1906 shakedown as among the severest earthquakes on the North American continent.

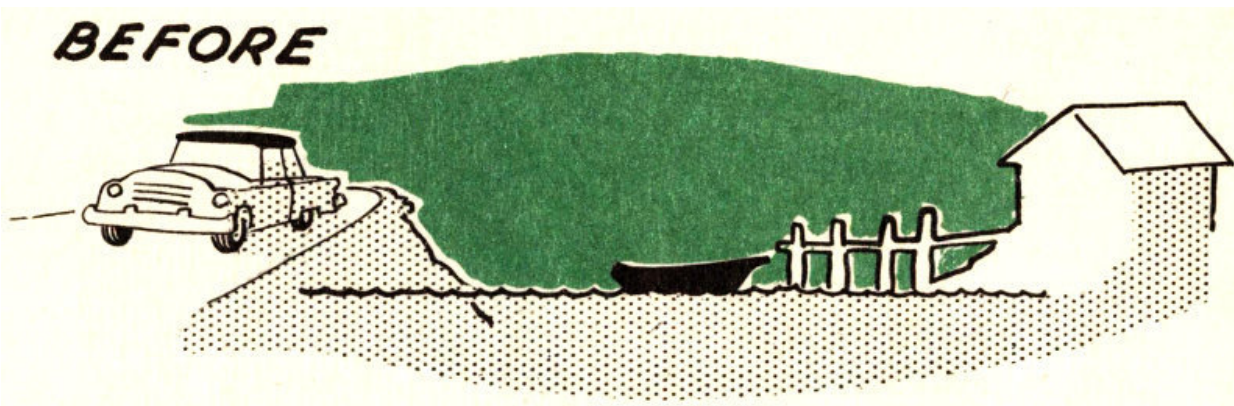
In seismic measurements, it rated 7.8 on the Richter Scale, as compared with San Francisco's 8.2.

It set up so called tidal waves, or seiches, on Hebgen Lake. There were at least three of these huge waves—20-ft. high—which overtopped the entire 721-ft. length of the dam by four feet. Eyewitness statements relate that the velocity of the tidal wave was so great that it caused the water literally to leap over the top of the dam. It filled the small generating plant with a 2 to 4 ft. deep layer of rocks.

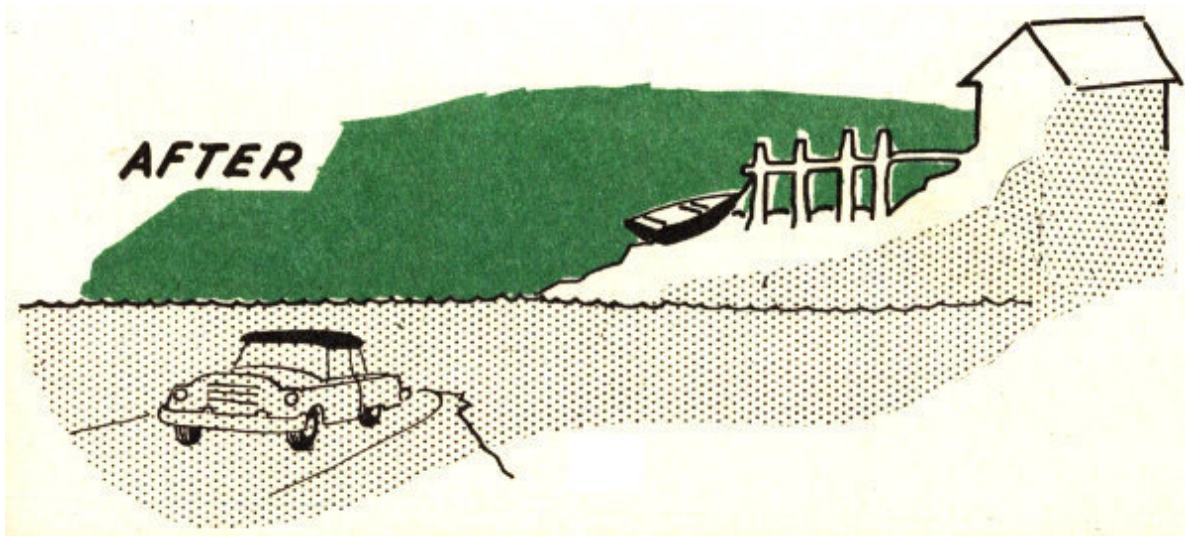
Although the dam stood, the quake caused several fractures in the core wall, one of which showed a 3- to 4-inch separation, and shattered the dam's concrete spillway.

The earthquake created three major faults, with displacement on the Red Canyon Fault running as much as 20 ft., which stacks up impressively alongside the 26-ft. maximum displacement resulting from San Francisco's quake. (These two earthquakes differed, however, in that the Montana displacement was vertical, while San Francisco's was horizontal.)

77



BEFORE



AFTER

According to the Society of Military Engineers, surveys from benchmarks outside the earthquake-affected areas show that the earth in the Hebgen Dam Quake area near Hebgen Dam has settled between eighteen and nineteen feet from its level before the quake.

It wasn't uniform, though. The quake caused tilting, which showed up in the way the north side of Hebgen Lake had sunk eight feet, while the south side of the lake, docks, boats, etc., were sticking eight feet out of the water.

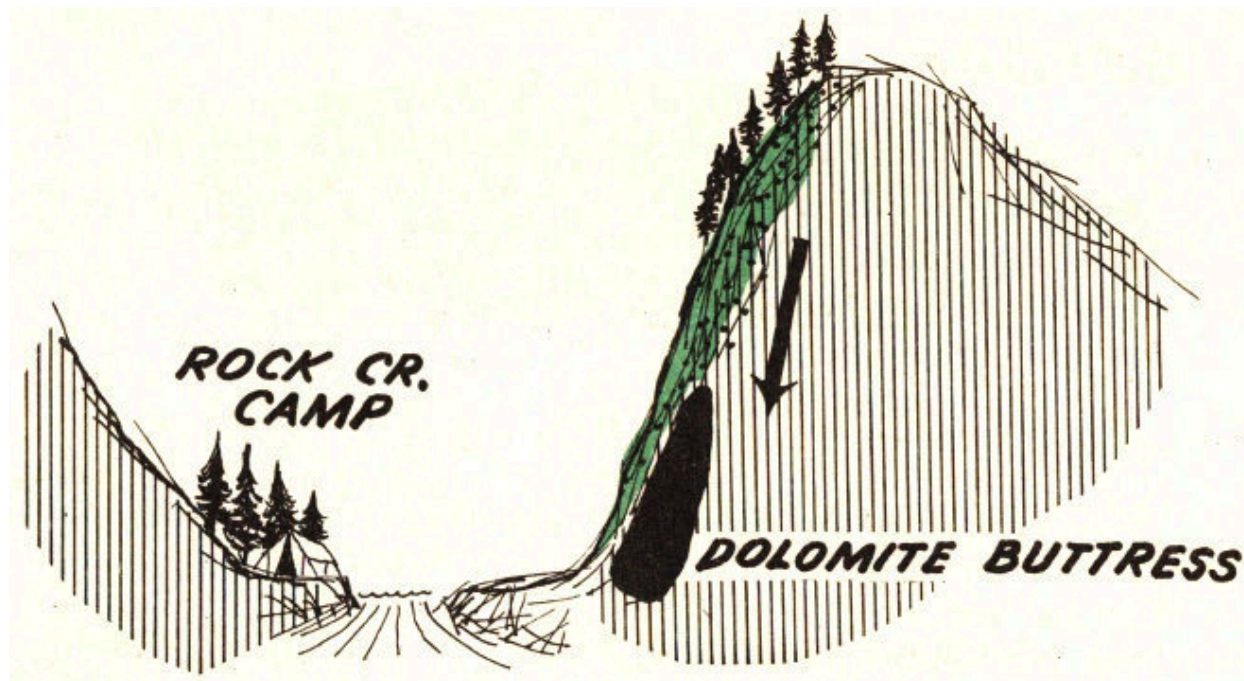
The quake also caused many sink, or more properly, blow holes. These phenomena are also known as sandspouts. Water, compressed and forced up and out by quake action washes out layers of sand sub strata. The overhead surface areas naturally drop into the hole, leaving a puzzling hunk of slumped ground—separate from the normal scarps—as big as 15 × 50 ft. in area.

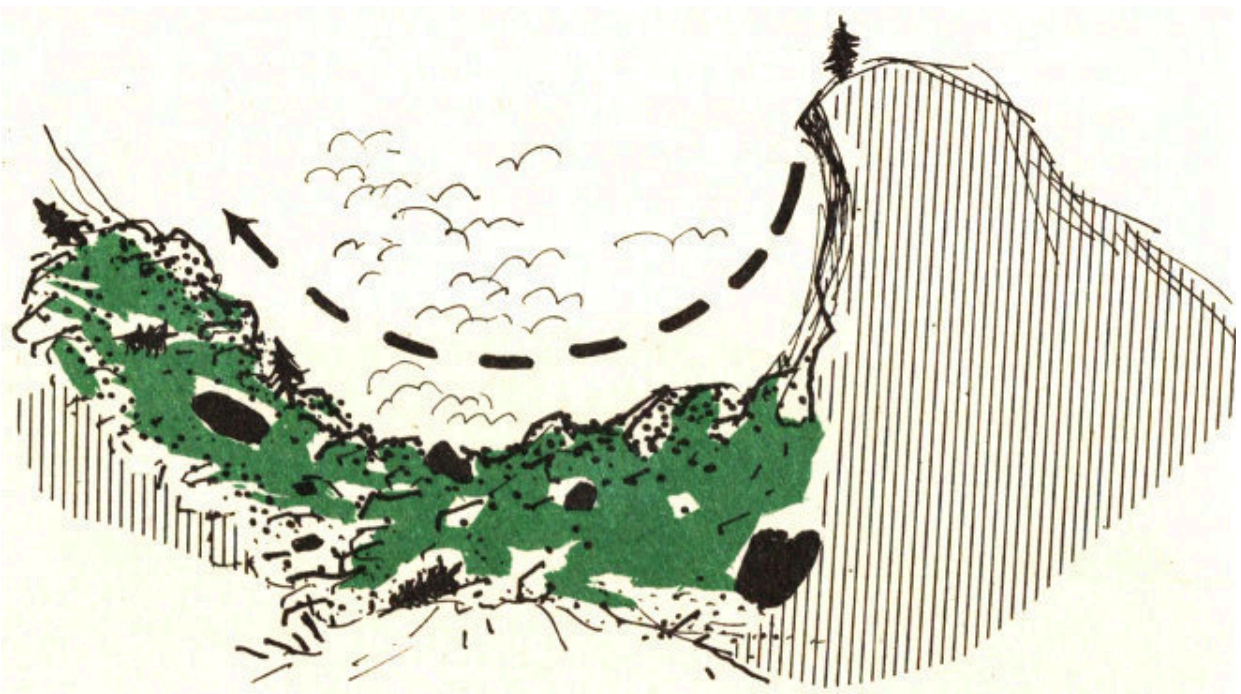
The Montana-Yellowstone quake sent seismographs jiggling as far away as New Zealand. It caused fluctuation of water level in wells as much as ten feet in nearby Idaho, a tenth of a foot in Hawaii, 3,200 miles away, and .01 ft. in Puerto Rico.

The huge concrete Hungry Horse Dam, near Columbia Falls, Montana, 250 miles NW of the quake area, showed measurable displacement as a result of

the quake. In remote Seattle, the diminished tremors were still strong enough to break loose the floating amphitheatre in Lake Washington.

78





But by far the most spectacular effect of Montana's Earthquake was the huge landslide at the mouth of the Madison Canyon.

At the site of the slide, a relatively strong and nearly vertical layer of dolomite rock supported a huge bank, or mountain, of comparatively unstable schist and kept it from sliding into the valley in the same way that a retaining wall keeps a hillside terrace from slipping downhill. The tremendous shock waves of the earthquake fractured this dolomite buttress, and some 43 million cubic yards, or 80 million tons of rock, timber, and other mountainside debris cascaded off the slope, hurtled into the canyon, and surged up the opposite side, carrying huge trees and house-sized boulders as if they were weightless, hollow toys.

When this huge mass whomped down onto the river bed, it forced out the water and air trapped underneath at hurricane velocity. The huge slide spurted mud, air, and water with such force as to send two-ton cars sailing through the air, and to grind others to suitcase thickness against the rocks.

All this happened in seconds.

It would take eight seconds for the mass at the top of the mountain to fall to the valley floor 1,200 ft. below. At the time it reached this point, the mass would be travelling 174 miles per hour. The time it took to zoom half-a-mile across the valley, up the opposite canyon wall, then split and flow three-quarters-of-a-mile up and down the valley (the slide lies one-and-a-half-miles-long in the valley), was less than thirty seconds!

The fact that timber from the face of the mountain is spread in relatively uniform fashion over the entire surface of the slide is interpreted to mean that there was little tumbling action—that the slide moved as a single, if shattered, mass.

One important scientific controversy has emerged from the earthquake. It relates to the time relationship, or sequence between the initial shock, the tidal waves, or seiches, how fast the huge quantities of water which overtopped the dam moved down the valley, and whether these slugs of water had rushed through the canyon in time to reach the site of the slide before the mountain fell.

The stretch of the Madison running through the canyon is fresh, fast water, but normally it takes up to two hours for an object to run the sparkling, 80 seven-mile, trout-rich stretch from Hebgen Dam to the mouth of the canyon. The big surges of water—the seiches overtopping the dam—would make it a lot faster.

There are two big, related questions.

Could the big surges of water reach the point of the slide soon enough?

And just how soon after the first shock did the mountain fall?



For the first couple of days after the quake, the theory persisted that the slide must have happened quite some time after the first shock—as late as 5:00 A. M., according to some theorists.

But, as the facts, and the testimony of folks trapped near the slide—the Osts, Fredericks, Smiths, and Mrs. Bennett—became available, it was apparent that the slide must have closely followed the initial shock. Even if you discount the disrupted time sense of people under stress—when a minute can seem like an hour, and vice versa—it's difficult to imagine that more than 20 minutes elapsed between the first shock and the slide.

According to one set of calculations, big waves could have swept from the dam to the slide site in 18 minutes or so.

81



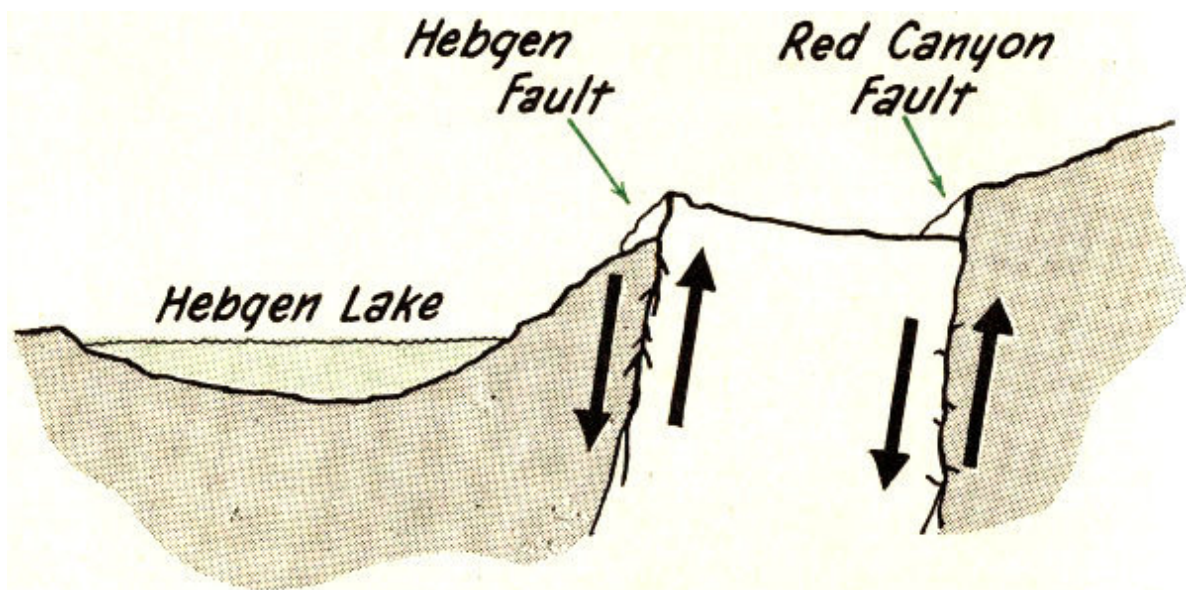
A man stands in the gap left by the earthquake-caused simple fault scarp.

(U. S. Geological Survey)



Something slipped here! Before the quake caused the ground drop, creating this magnificent scarp, the ground surface shown here was continuous.
(U. S. Forest Service)

82



Hebgen Lake
Hebgen Fault
Red Canyon Fault

Although the quake caused much settling of the earth packed against the downstream side of Hebgen Dam's concrete core, the relatively slight displacement of the sod cover is interpreted to mean that all three tidal waves passed over the dam before this earth subsided and separated from the core. Thus the water would have begun its race down the valley before the heavy earth-settling shocks hit the dam area.

Those who support the high-water-at-the-moment-of-the-slide theory point to the great volume of water damage way below the slide.

If the slide had come first, it would have dammed off the tidal waves, and prevented such damage. They feel there just wasn't enough water in the river bed's normal content to cause the water damage done both upstream and downstream by the slide. And they argue, the mud and dust in the composition of the slide would have taken up most of the water normally found in the reach of the river buried under the slide.

There's further evidence in the numerous fish found high and dry on the flat along the river bank several feet higher than the streambed. Most of them were small, catfish-like chubs. There were numerous trout, and one 18-inch carp. There is no place in the river below the pool at the toe of the dam where carp would likely be found.

Also, there was further confirmation in the fact that three of the especially made 11-inch squared timbers, eight and a half feet long, with notched ends and two U-bolts used as stop-logs in the Hebgen Dam spillway were found below the slide. Some shadow was cast on this as absolute confirmation by the Montana Power Co.'s explanation that stop-logs have been lost from time to time before the quake.

83

Those who, in spite of such evidence, oppose the theory that the high water reached the slide area first just don't feel that the water could have made it all the way down the canyon in so short a time. They feel that it would have taken at least 40 minutes for the big waves to traverse the seven miles. They

have some support in L. D. Smith's testimony that in driving down from Beaver Creek to Rock Creek right after the shocks, he saw no such waves.

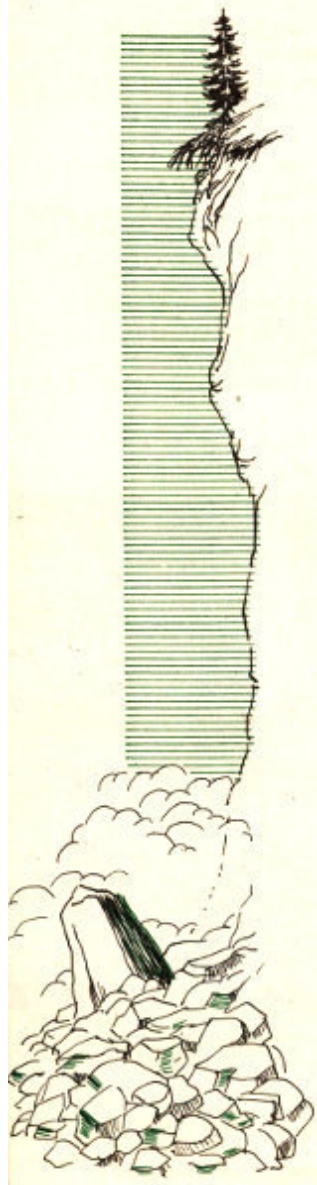
At any rate, this is one argument that geologists and hydrologists will be batting around for a long time.



This view of the Madison Canyon slide gives the feeling of the up-canyon and down-canyon flow of the 80-million-ton mass of rock and debris. (Montana Highway Commission)

NOW YOU CAN SEE

It's unusual when an event so spectacular as the Montana-Yellowstone Earthquake doesn't produce some exploitable possibilities, and this one did. The month after this August 17, 1959 series of quakes, the U. S. Forest Service, which is proprietor of the vast and tumultuous real estate on which the major portion of the immediate quake action happened, announced that is was underway with plans to set up a Geological Area to help visitors get to earthquake interest points and to understand the tremendous earth forces which operated here.

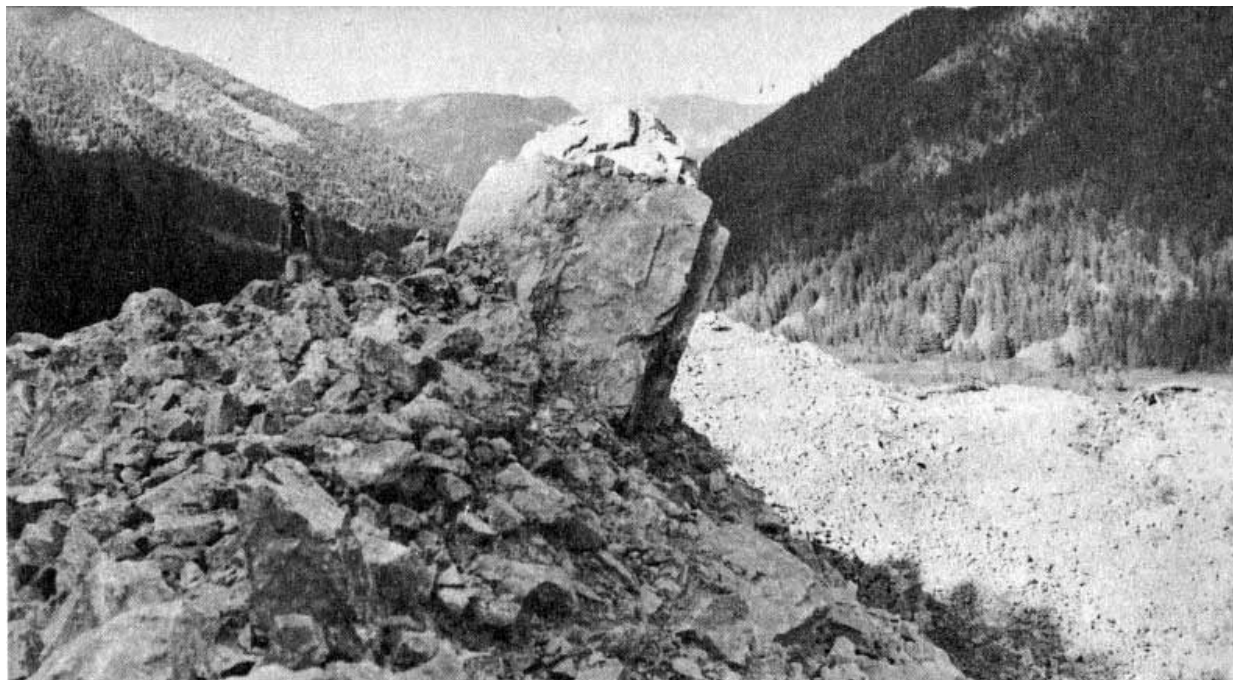


They held the inaugural of the Madison River Canyon Earthquake Area—the first of its kind anywhere—on August 17, 1960, the first anniversary of the quake. Relatives of the 28 quake victims sat on the gigantic slide as they unveiled the bronze memorial plaque mounted on the huge dolomite boulder which had floated across the valley atop the surging debris.

This awesome and fascinating earthquake area quickly became one of the region's top tourist attractions, with close to half-a-million visitors in attendance during each of the first two post-quake summers, in spite of miserable to nearly impassable access roads. This popularity is especially

fitting because the quake that's on display here was essentially a "tourist earthquake". It happened in the scenic mountain area which draws a brisk vacation traffic from all over the U. S. and Canada, during the height of the tourist season. And those who went through the adventure, the thrills, the terror, the heroes and the helped, the survivors, and the casualties, could nearly all be classed as tourists.

85



On this huge boulder, atop the slide, the U. S. Forest Service erected a plaque in memorial to the victims of the Montana-Yellowstone Earthquake. (U. S. Forest Service)

Superb trout fishing has always been one of the area's most important features, and, understandably, there was much post-quake concern as to how this would be affected. While Hebgen Lake was drained to repair the quake-damaged Hebgen Dam, Montana's Fish and Game Department poisoned the trash fish and stocked the refilled lake with millions of rainbow trout running in size from fingerlings to 9-inchers. Today both Hebgen and Quake Lake—formed by the damming action of the slide at the mouth of Madison Canyon, afford top fishing, either from shore or from boat. Quake Lake has a made-to-order launching ramp at Cabin Creek, where the flooded-out road runs right into the lake.

In spite of the concern by fish biologists that silting from the slide would take the edge off fishing on the Madison below the slide area, it kept right on providing fishermen the top-notch action that had long earned its reputation as a blue-ribbon trout stream that compares with fishing anywhere in the world.

Today there are excellent roads to and through the Earthquake Area. Route 287, south from Ennis, leads directly to the huge slide at the mouth of Madison Canyon. Here the Forest Service has built a surfaced road up onto the slide. On top is the best vantage point to view the whole panorama 86 of the mountain fall—where it dropped from, how in a matter of seconds 80-million tons of rock cut off the valley, the sparkling blue lake it created, and the open stretch of the Madison below the canyon. Besides, on the slide there are interpretive exhibits, and the huge monolith bearing the plaque to the quake dead, 21 of whom still lie somewhere beneath this mammoth pile of rock.

The relocated route runs eastward down the slide, along Quake Lake, and through Madison Canyon. Several Forest Service people staff the formally designated Earthquake Area during the summer season to help explain and interpret what happened here. Definite plans for the area include a formal visitor center, and at least one first-class campground in the slide-lake-canyon area.

Hebgen Dam, the dam that held, straddles the upper end of Madison Canyon. The road from here along Hebgen Lake to the Duck Creek Y has been much improved over its pre-quake status.

The Quake Area is just as easily approachable from West Yellowstone by taking 191 north 10 miles to the Duck Creek Y, and then driving west along Hebgen Lake. Near the Y, the big fault runs close to the road, through the Culligan ranch, etc.

The magnificent Reynolds Pass road, which runs south from its junction with the Madison Canyon road three miles west of the slide, has become an important new route to the earthquake area. The morning after the slide, highway crews were at work on this alternate route, which for two years substituted for the blocked, flooded, and destroyed road through the Madison

Canyon and along the north shore of Hebgen Lake while the regular route between Ennis and West Yellowstone remained blocked. With its exciting mountain backdrop, this new, improved road provides an enjoyable alternative which should be included in any circle tour of quake features.



In the spring of 1959, as he tells it, Lemuel Garrison, Superintendent of Yellowstone National Park, looked at some bids for new housing in the Park which included extra steel as a protection against the possibility of earthquakes.

“Heck,” he said, “We’re not in an earthquake area.”

Today, Yellowstone’s famous earthquake has become an important addition to its already fabulous attractions. The Park took the quake in its stride. By June 1, 1960, in spite of road damage of \$2,600,000, and building damage of \$1,700,000 resulting from the quake, Yellowstone Park, its roads, and other facilities were ready for its normal summer rush.

87

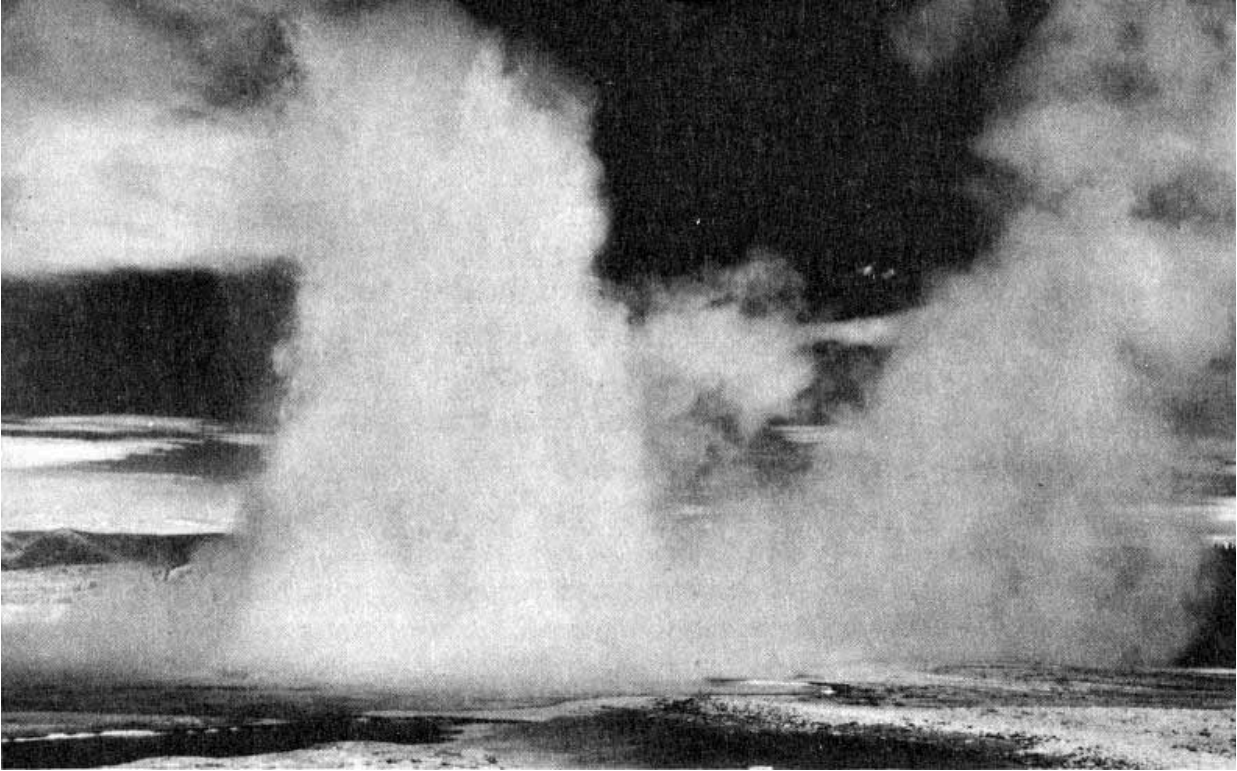
In clearing a slide which blocked the road near Firehole Falls, south of Madison Junction, the road crew discovered one near casualty—a bear. The bruin had evidently sought shelter in a hollow below the road shoulder, and became trapped when the slide closed his exit. It was several days after the quake when the crew heard the bear’s attempts to crawl out of his artificial cave. They lowered a tree trunk, still bearing branches, into the hole and retreated while the bear scrambled out.

Word of the quake plus the initial belief that the epicenters of seven of the eleven major shocks were located in the famous Firehole Basin caused widespread anxiety as to whether the tremendous forces loosed might have interfered with nature’s intricate underground plumbing which keeps the geysers, hot pools, and mud pots spouting, burbling and burping.

Studies by a horde of seismologists, geologists and other earth scientists who swarmed into the Firehole Basin in the months after the quake show that during the night of August 17 the hot spring activity in this area changed more than during the 87 years since a park was created out of the mysterious, steaming country which had been known as “Coulter’s Hell.” The scientists termed these changes as “profound and far reaching.” These changes in thermal features are, and will be, in years to come, tremendously interesting.

The majority of these changes came with, or just after, the initial quakes. The earthquake acted as a trigger to start eruption in hundreds of springs, nearly half of them erupting for the first time in their known histories. The whole place blew, then subsided.

There was considerable juggling of the intervals and playing times of some of the better-known geysers. Great Fountain, Riverside, Daisy, Castle, and Oblong shortened the length of their eruption intervals, but they play nearly twice as frequently as they did prior to the quake. Sapphire, a minor geyser, became a major geyser, but has subsided to a status somewhere in between. Clepsydra Geyser went frantically wild, and has erupted continuously since the quake.



The quake gave Yellowstone Park's famous Morning Geyser in lower Geyser Basin a shot in the arm. Normally erupting once a day, after the quake it blasted off every four hours.

(National Park Service)

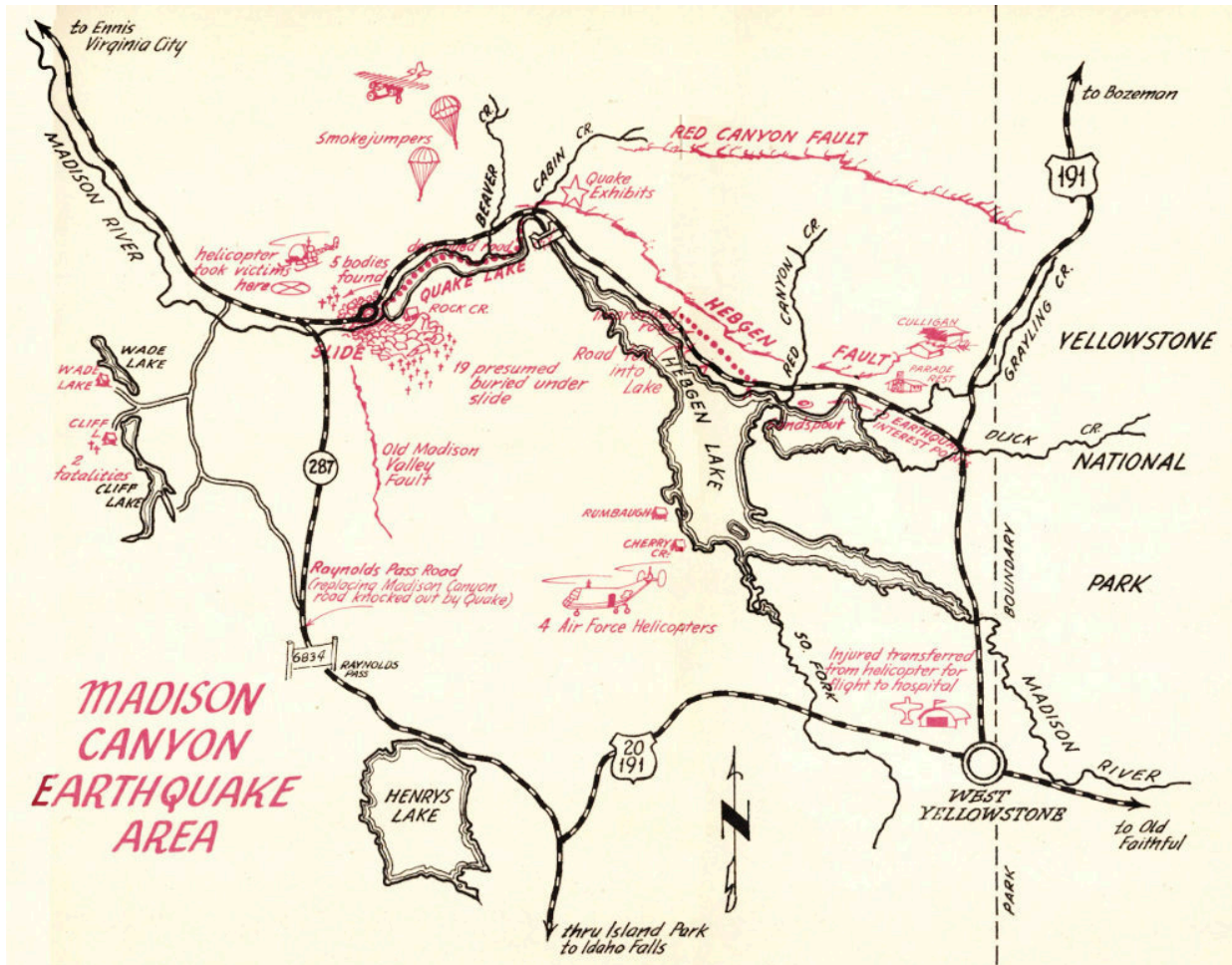
Steady Geyser just up and quit. So did Grand Geyser. Giantess Geyser, located just across the river from Old Faithful, habitually shook the ground in the vicinity every time it erupted. Right after the quake it blasted off and kept blowing for a continuous 100 hours, instead of its usual 30-hour run. The Fountain Paint Pots became so active, and spread so, that they took over what used to be an asphalt-paved parking area.

In the midst of these changing patterns, Old Faithful goes on in much the same way, except for perhaps a slight increased interval between blasts. Studies of 14,317 eruptions were clocked, with an average interval of 63.8 minutes. The shortest interval was 33 minutes, in 1948-51, the longest is 93 minutes, measured in '55.

But none of these changes are static. Just when Grand Geyser, having been dormant for five months, was considered dead, it moved into a sporadic blasting phase.

These quake-caused fluctuations in thermal features, plus strong curiosity as to the earthquake's effects in the surrounding area, Sup't. Lon Garrison feels, will make Yellowstone's post-quake years bigger than ever.

89



MADISON CANYON EARTHQUAKE AREA

QUAKE LAKE

Smokeyjumpers

SLIDE

destroyed road

5 bodies found

19 presumed buried under slide

helicopter took victims here

Quake Exhibits

Old Madison Valley Fault

RAYNOLDS PASS

Raynolds Pass Road (replacing Madison Canyon road knocked out by Quake)

CLIFF LAKE

2 fatalities

RED CANYON FAULT

HEBGEN LAKE

HEBGEN FAULT

Road fell into Lake

Improvised road

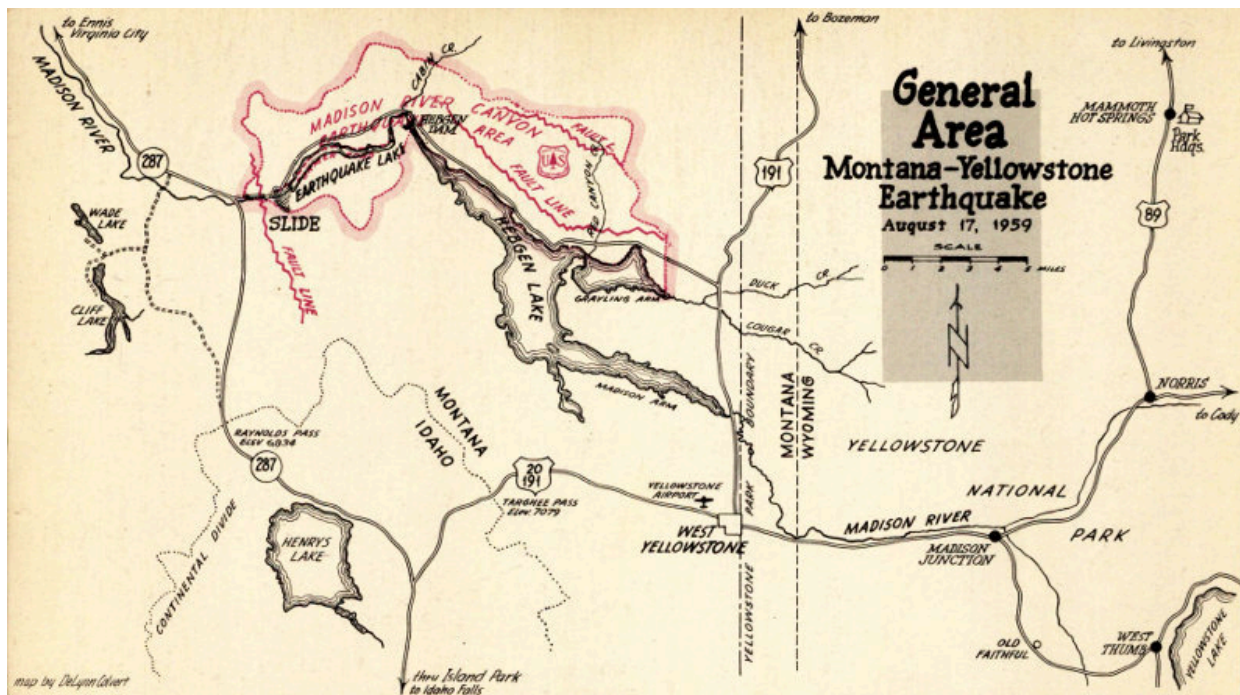
Sandspout

4 Air Force Helicopters

WEST YELLOWSTONE

Injured transferred from helicopter for flight to hospital

90



General Area

Montana-Yellowstone Earthquake

August 17, 1959

91

MONTANA-YELLOWSTONE EARTHQUAKE



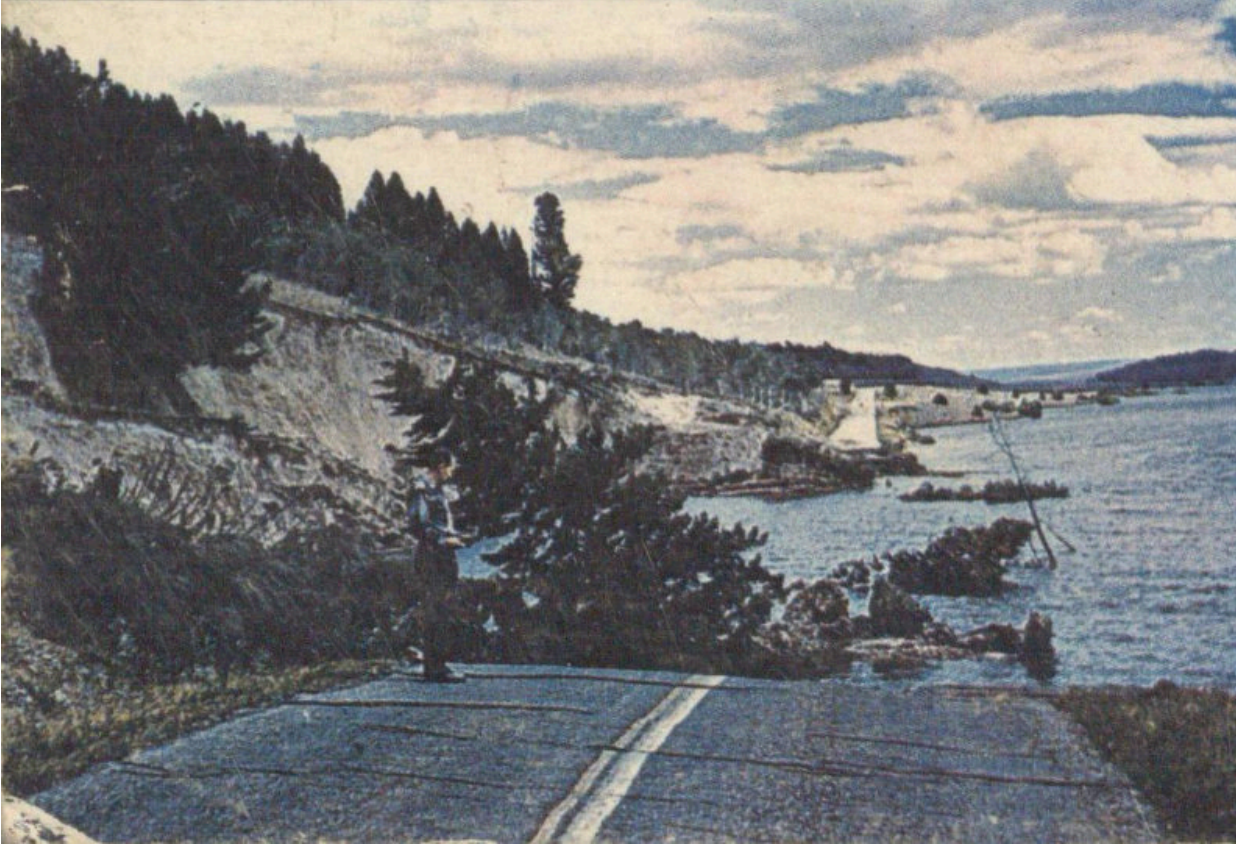
The Slide

(U. S. Forest Svc.)



This was level ground before the quake made this 20-ft. scarp.

(Mont. Hwy. Comm.)



Road missing after quake.

(Montana Power)



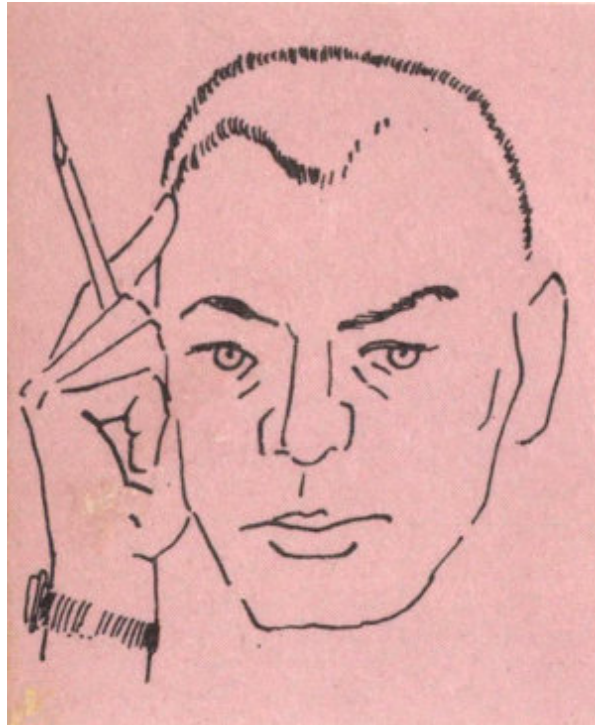
These massive rocks blocked the road at Yellowstone Park's Golden Gate after the Earthquake.
(Natl. Park Svc.)



Shook-up road after quake.

(USFS)

ABOUT THE AUTHOR



Ed Christopherson was a professional author and magazine writer whose articles about Montana, the Northwest, and other subjects appear in *The Saturday Evening Post*, *Holiday*, *This Week Magazine*, *Mademoiselle*, *Reader's Digest*, *The New York Times*, *Congressional Record*, etc.

Born in Ohio, he began his writing career in New York. His introduction to the Mountain Northwest came through a season as a Forest Service Smokejumper. After several years in New York, he picked exciting and scenic Western Montana as the center of his regional writing activities.

Christopherson went to West Yellowstone (they called it "Shookville") the day after the quake. He got first hand accounts from survivors there, and in Ennis, flew and walked over the slide and elsewhere in the quake area, and

since has spent months researching and correlating what turned out to be
“The Night The Mountain Fell.”

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